Chapter 11

VOLTAGE REGULATORS

Review of Rectifier Circuit

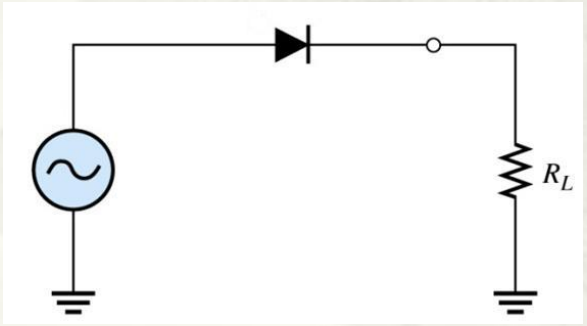
Three elements of a Power Supply:

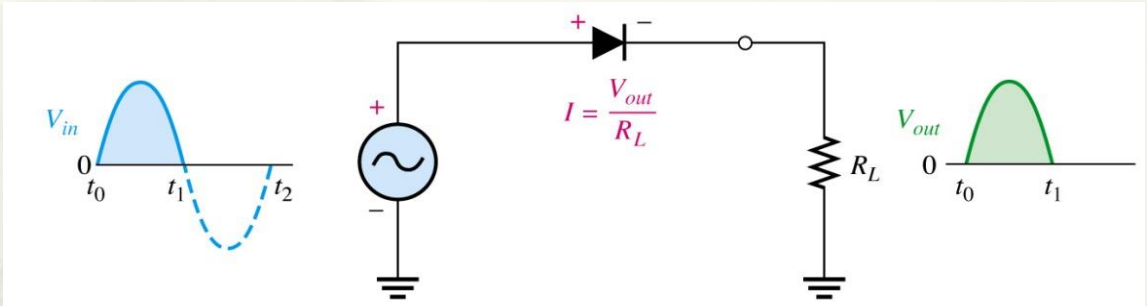
Rectifier circuit, Filter, Regulator

- Half-Wave Rectifiers
- Full-Wave Rectifiers
- Bridge Rectifiers

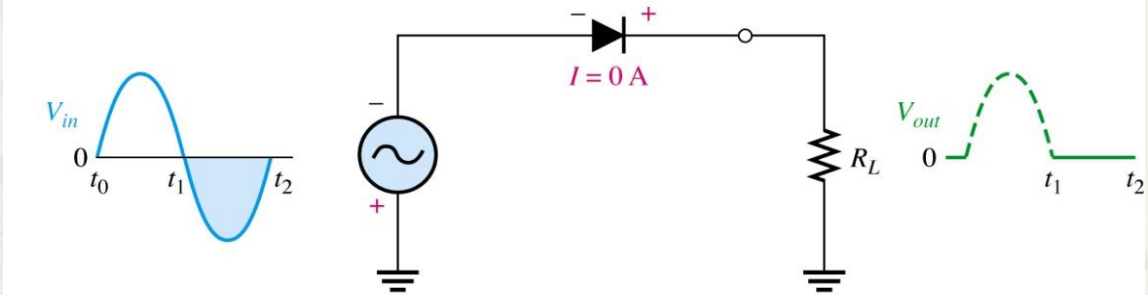
Half-Wave Rectifier

- **Rectifier**
  - convert ac into pulsating dc.
- **Half-wave rectifier**
  - $V_{p(out)} = V_{p(in)} - 0.7\text{V}$
- **Rectifier Diode**

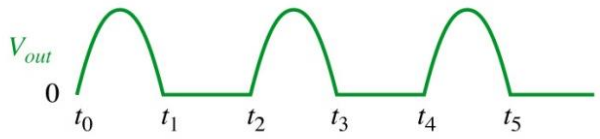




(a) Operation during positive alternation of the input voltage, diode conducts.



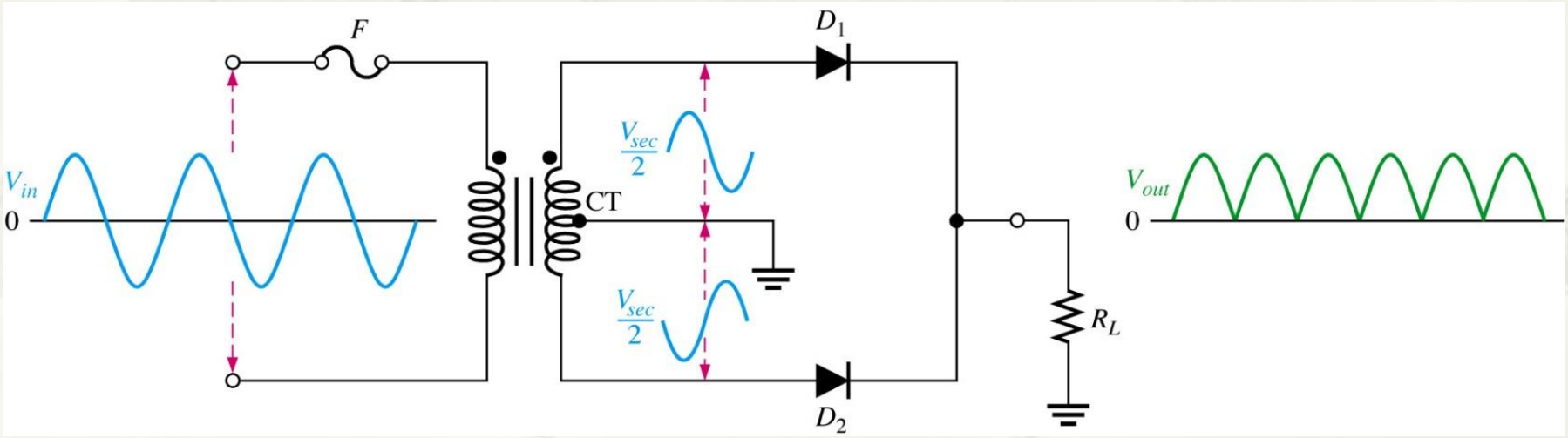
(b) Operation during negative alternation of the input voltage. Diode does not conduct; therefore, the output voltage is zero.



(c) Half-wave output voltage for three input cycles

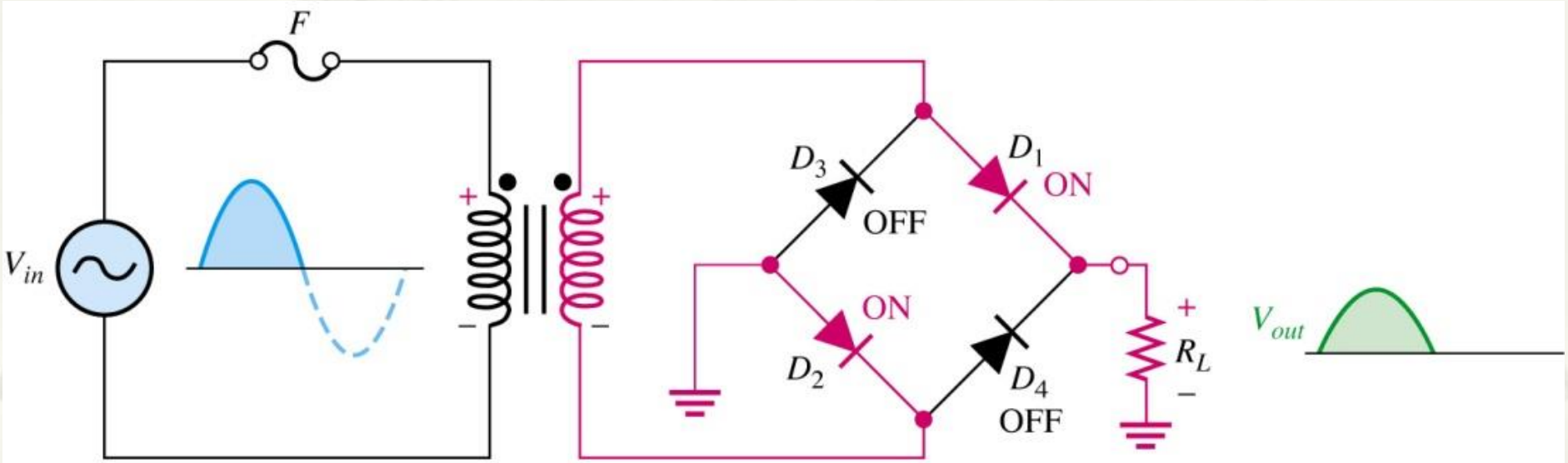
Full-Wave Rectifier

- **Full-wave rectifier**
  - allow unidirectional current to the load during the entire input cycle.
  - The dc output voltage pulsates every half-cycle of the input.
- **Center-tapped (CT) full-wave rectifier**



Bridge Rectifier

- **Bridge Rectifier**
  - Does not require a center-tapped transformer.
  - The four diodes are available in a single package.

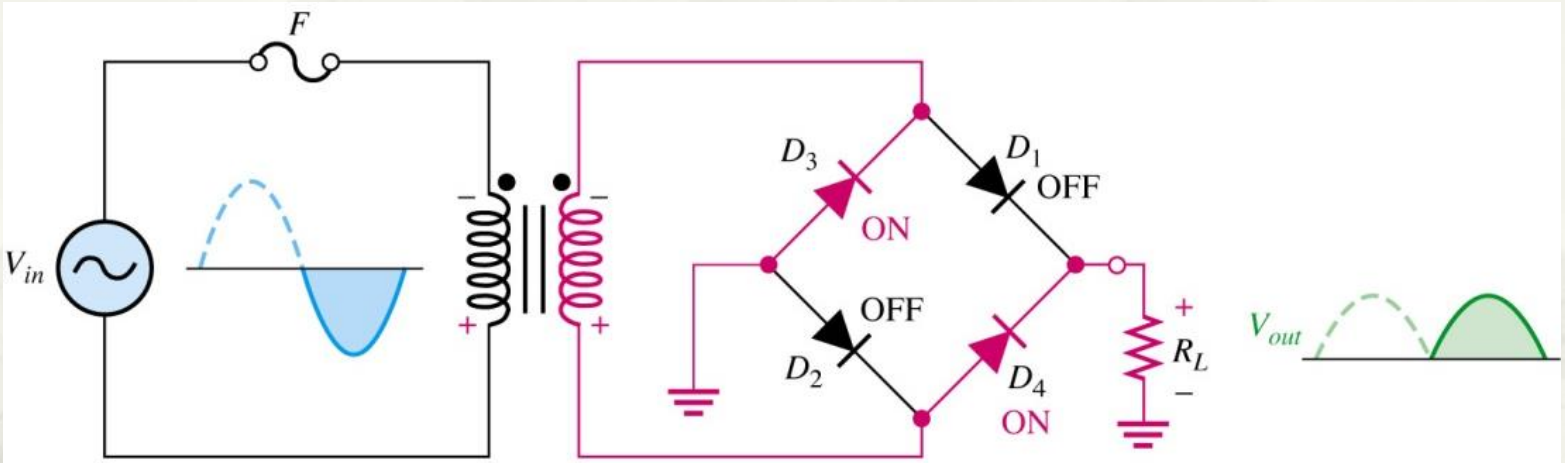


(a) During positive half-cycle of the input, D_1 and D_2 are forward-biased and conduct current. D_3 and D_4 are reverse-biased.

Bridge Rectifier

Bridge Rectifier

- Does not require a center-tapped transformer.
- The four diodes are available in a single package.



(b) During negative half-cycle of the input, D_3 and D_4 are forward-biased and conduct current. D_1 and D_2 are reverse-biased.

- Actually, Output Voltage: $V_{p(out)} = V_{sec} - 1.4\text{V}$
- Peak Inverse Voltage: $\text{PIV} = V_{p(out)}$

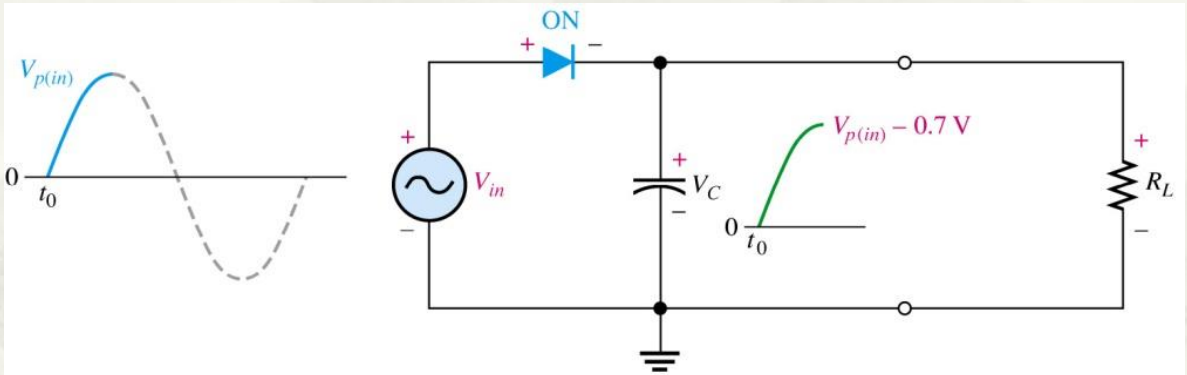
FILTERS AND REGULATORS

The function of a power supply is to convert the standard power frequency ac power line voltage to nearly constant dc.

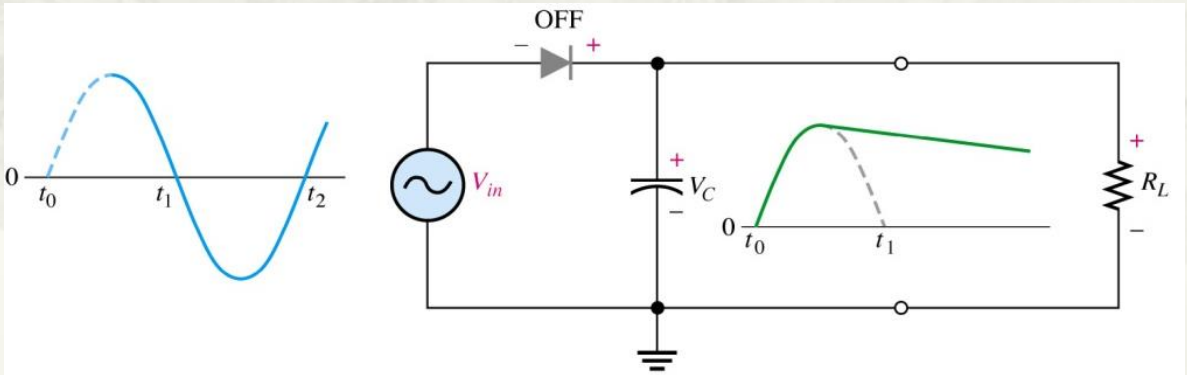
- Capacitor-Input Filter
- Zener Regulators
- IC Regulators
- Percent Regulation

Capacitor-Input Filter

- **Capacitor-input filter** is the least expensive and most widely used power supply filter



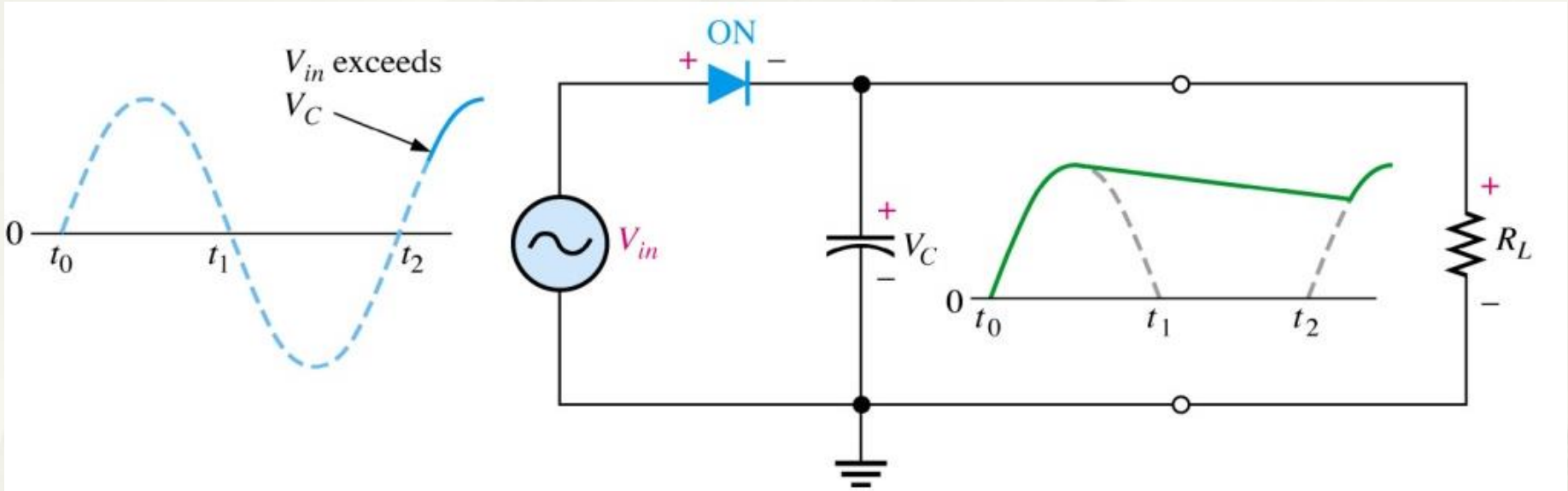
(a) Initial charging of capacitor (diode is forward-biased) happens only once when power is turned on.



(b) The capacitor discharges through R_L after peak of positive alternation when the diode is reverse-biased. This discharging occurs during the portion of the input voltage indicated by the solid blue curve.

Capacitor-Input Filter

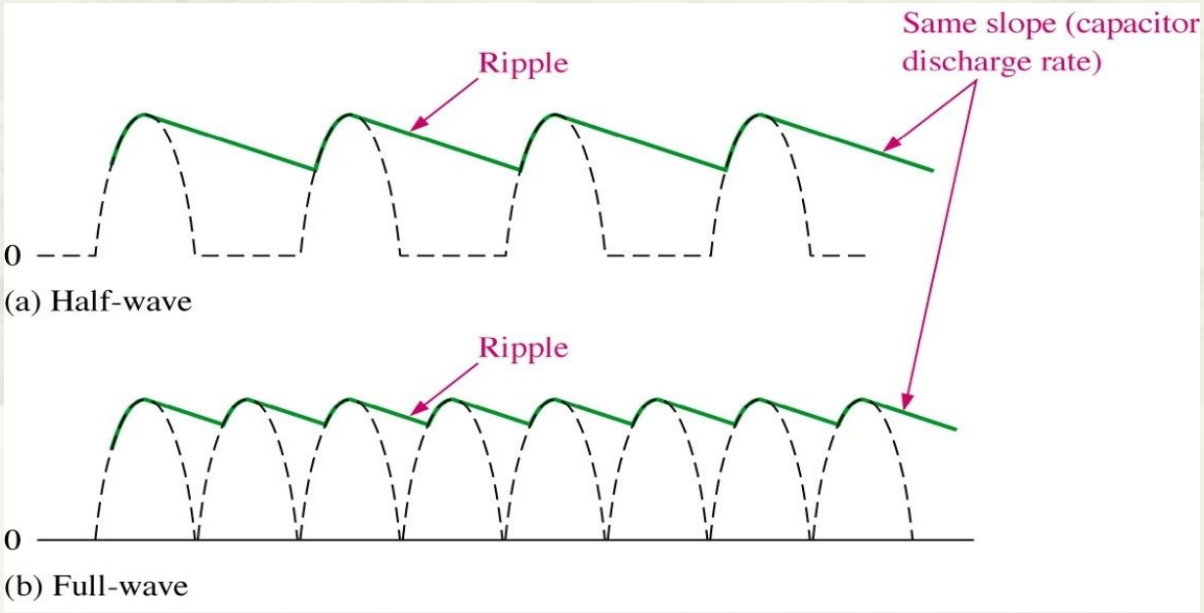
- **Capacitor-input filter** is the least expensive and most widely used power supply filter



(c) The capacitor charges back to peak of input when the diode becomes forward-biased. This charging occurs during the portion of the input voltage indicated by the solid blue curve. Notice that the diode is not forward-biased on the second cycle until the capacitor voltage is overcome.

Capacitor-Input Filter

- **Ripple Voltage**
  - The variation in the capacitor voltage due to the charging and discharging is called **ripple voltage**.
  - When filtered, the full-wave rectified voltage has less ripple voltage than does a half-wave voltage for the same load resistance and filter capacitor values.



CHAPTER OUTLINE

11-1 Voltage Regulation

11-2 Linear Series Regulators

11-3 Linear Shunt Regulators

11-4 Switching Regulators

11-5 Integrated Circuit Voltage Regulators

11-1 Voltage Regulation

- Line Regulation
- Load Regulation

Line Regulation

Line Regulation

- Line regulation is a measure of the ability of a power supply to maintain a constant output for changes in the input voltage.
- $Line\ regulation = \left(\frac{\Delta V_{\mathrm{OUT}}}{\Delta V_{\mathrm{IN}}}\right) 100\%$
- Unit: mV/V or a pure number
- $Line\ regulation = \left(\frac{\Delta V_{\mathrm{OUT}}/V_{\mathrm{OUT}}}{\Delta V_{\mathrm{IN}}}\right) 100\%$
- Unit: %/mV or %/V

Example 11-1

Problem

When the input to a particular voltage regulator decreases by 5 V, the output decreases by 0.25 V. The nominal output is 15 V. Determine the line regulation expressed as a percentage and in units of %/V.

Solution

From Equation 11–1, the percent line regulation is

$$\text{Line regulation} = \left(\frac{\Delta V_{OUT}}{\Delta V_{IN}}\right)100\% = \left(\frac{0.25\text{ V}}{5\text{ V}}\right)100\% = \mathbf{5\%}$$

From Equation 11–2, the percent line regulation is

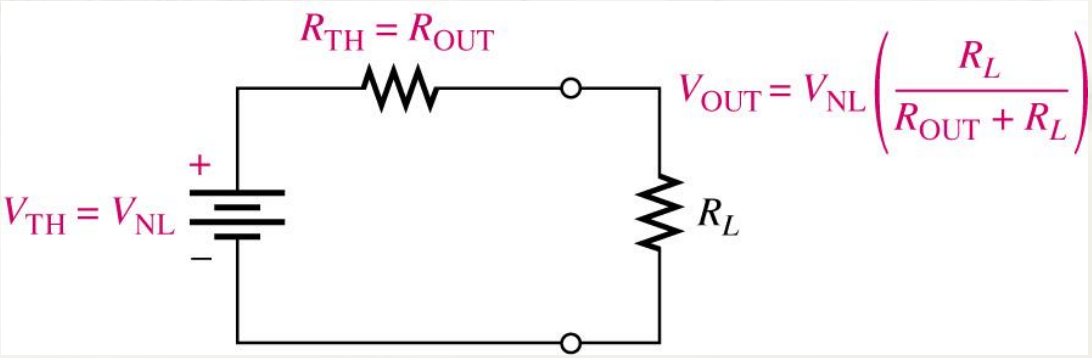
$$\text{Line regulation} = \left(\frac{\Delta V_{OUT}/V_{OUT}}{\Delta V_{IN}}\right)100\% = \left(\frac{0.25\text{ V}/15\text{ V}}{5\text{ V}}\right)100\% = \mathbf{0.33\ \%/V}$$

Load Regulation

Load Regulation

- The percent load regulation specifies how much change occurs in the output voltage over a certain range of load current values.
- $Load\ regulation = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right) 100\%$
- Unit: mV/V or a pure number
- Use output resistance of a power supply (R_{OUT})
- $Load\ regulation = \left(\frac{R_{OUT}}{R_{FL}}\right) 100\%$

V_{FL}—Full load
V_{NL}—No load



Example 11-2

Problem

A certain voltage regulator has a +12.1 V output when there is no load ($I_L = 0$) and has a rated output current of 200 mA. With maximum current, the output voltage drops to +12.0 V. Determine the percentage load regulation and find the percent load regulation per mA change in load current.

Solution

The no-load output voltage is

$$V_{NL} = 12.1 \text{ V}$$

The full-load output voltage is

$$V_{FL} = 12.0 \text{ V}$$

The percent load regulation is

$$\text{Load regulation} = \left(\frac{V_{NL} - V_{FL}}{V_{FL}}\right)100\% = \left(\frac{12.1 \text{ V} - 12.0 \text{ V}}{12.0 \text{ V}}\right)100\% = \mathbf{0.83\%}$$

The load regulation can also be expressed as

$$\text{Load regulation} = \frac{0.83\%}{200 \text{ mA}} = \mathbf{0.0042 \ \%/mA}$$

11-2 Linear Series Regulators

- Voltage References
- Regulating Action

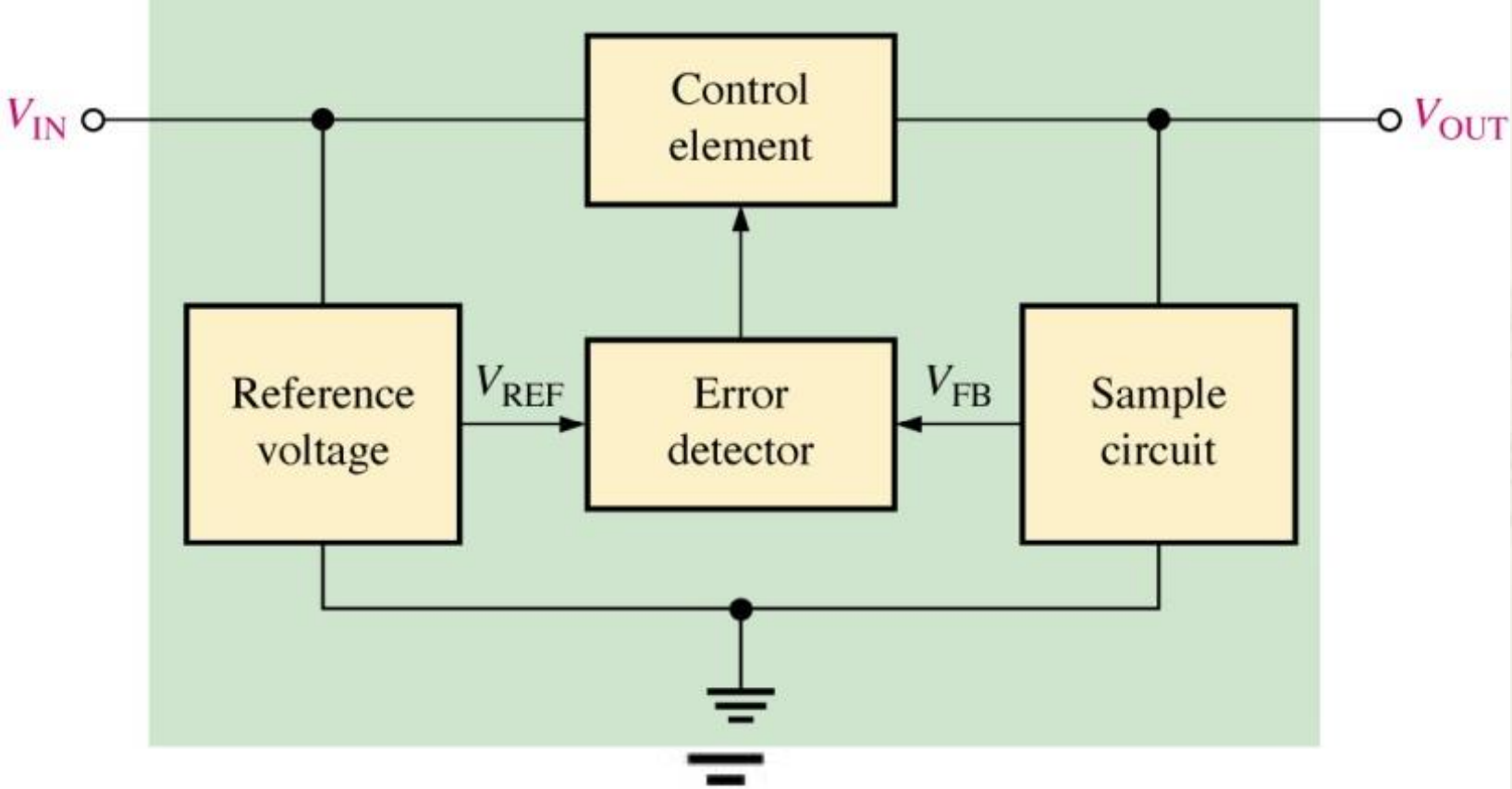
Linear Regulators



- **Linear regulator**
  - A linear regulator is a voltage regulator in which the control element operates in the linear region.

Linear Series Regulators

- **Linear Series Regulator**
  - The control element is in series with the load.



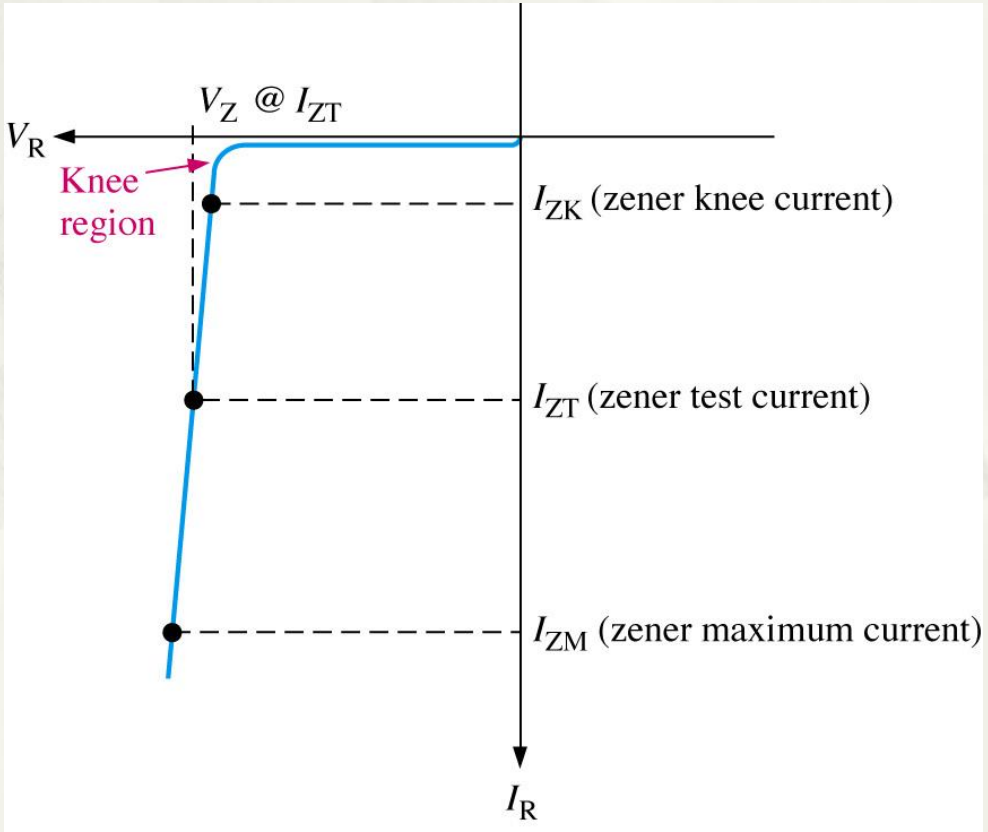
Voltage References

- **Voltage reference**
  - The ability of a voltage regulator to provide a constant output is dependent on the stability of a voltage reference to maintain a constant voltage for any change in temperature or other condition.
- **Zener diode reference**
  - Noisy
  - Temperature drift

Zener Regulators

Zener Regulator

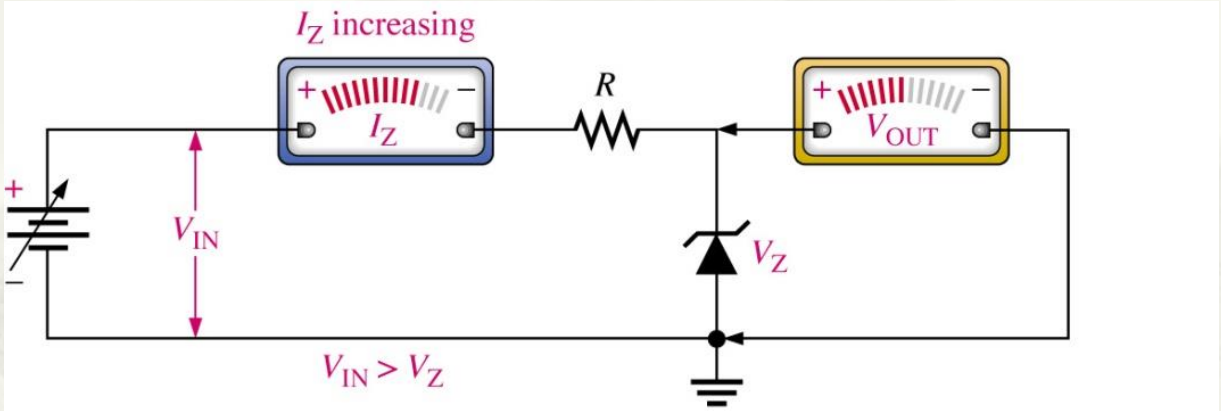
- A **zener regulator** is a special-purpose diode designed to conduct in the reverse direction.
- Precise breakdown voltage, controlled at the time of manufacture.



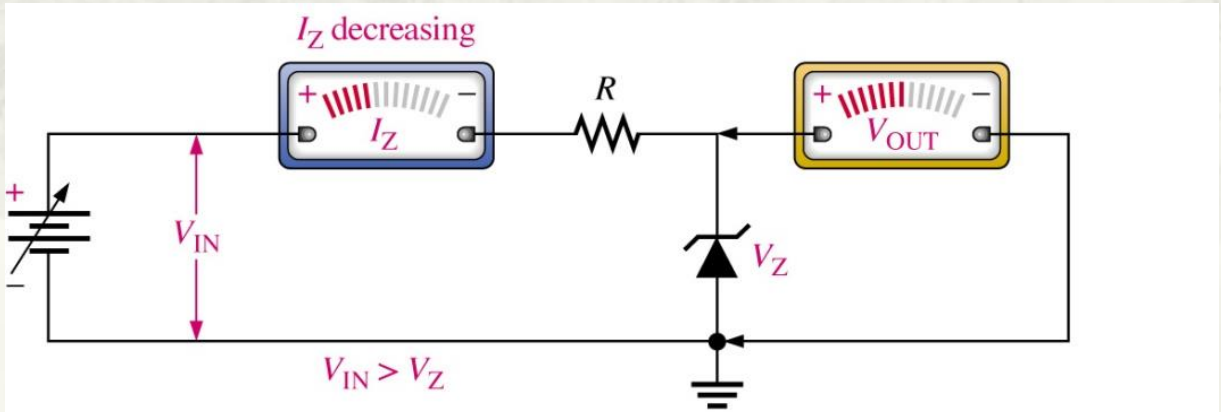
Zener Regulators

Voltage Regulation

- Regulate a varying dc input voltage to keep a constant level.



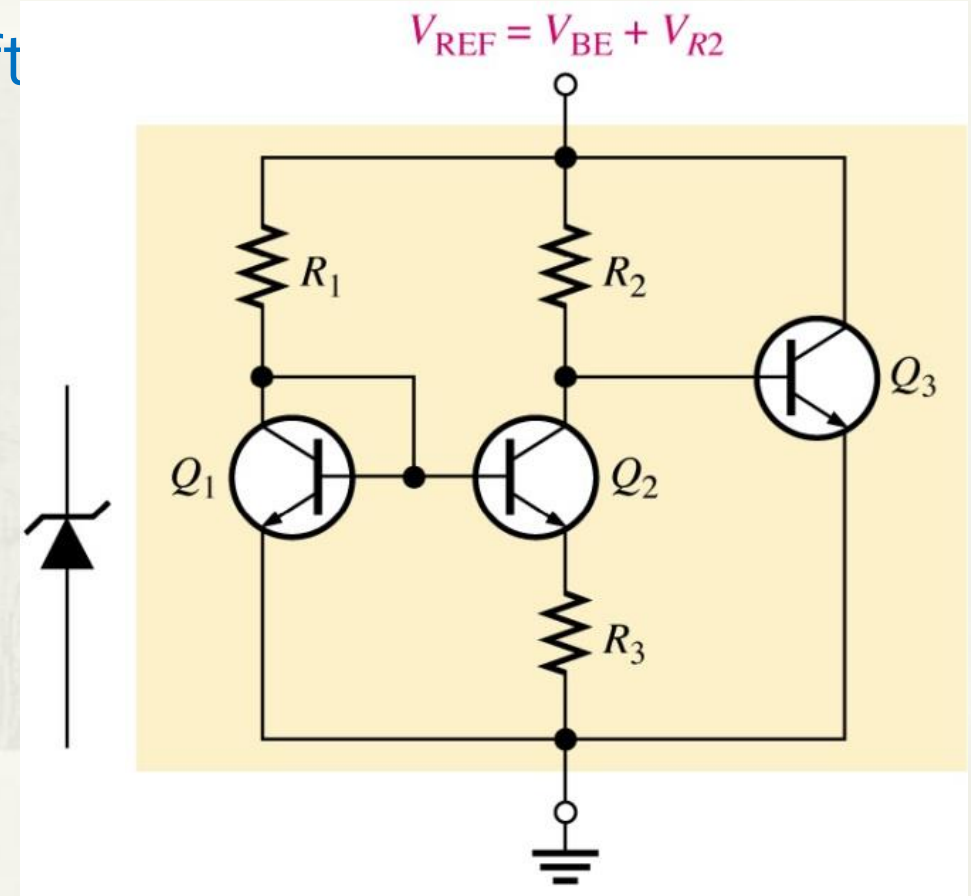
(a) As the input voltage increases, the output voltage remains constant ($I_{ZK} < I_Z < I_{ZM}$).



(b) As the input voltage decreases, the output voltage remains constant ($I_{ZK} < I_Z < I_{ZM}$).

Voltage References

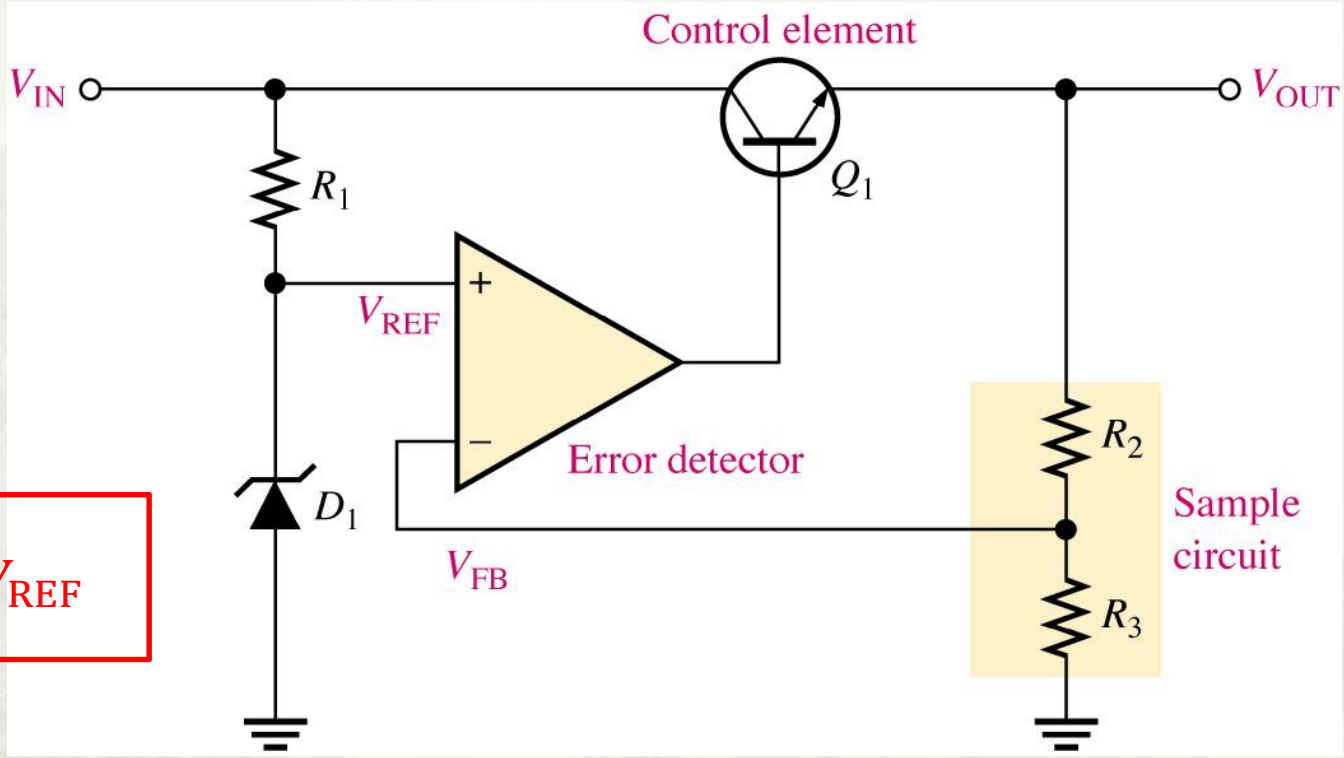
- **Zener diode IC**
  - Very low temperature drift
- **Bandgap reference**
  - It is designed so that positive and negative temperature coefficients cancel, producing a reference voltage with almost no temperature coefficient.



Voltage reference IC TL431

Regulating Action

- **Basic op-amp series regulator circuit**

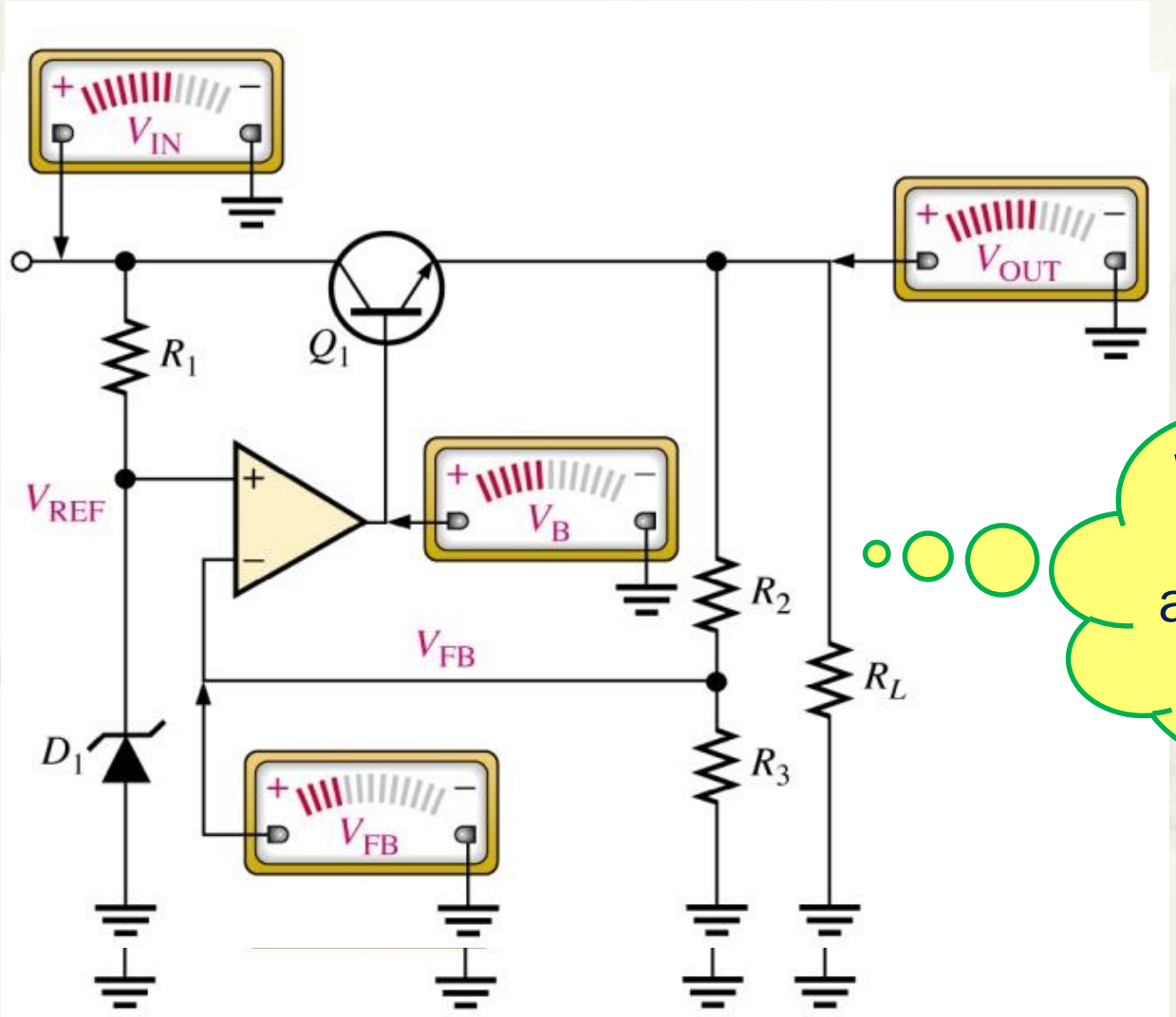


$$V_{OUT} = \left(1 + \frac{R_2}{R_3}\right) V_{REF}$$

- ✓ The output voltage is relatively independent of the input voltage, regulation is achieved.
- ✓ *The input voltage and load current should be within specific limits.*

Regulating Action

- **Illustration of series regulator action**

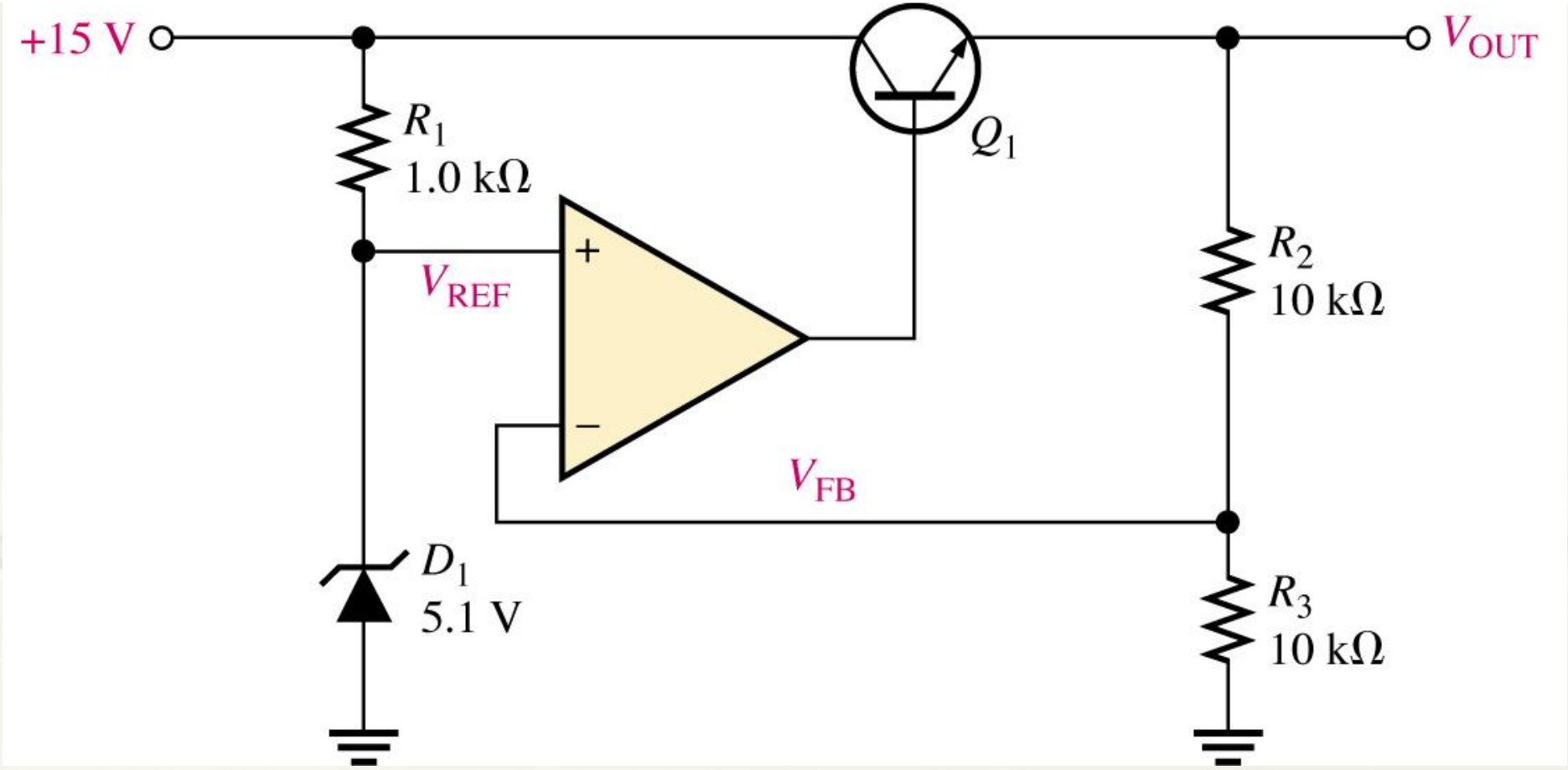


What is the range of output voltage and output current for this circuit?

Example 11-3

Problem

Determine the output voltage for the regulator in Figure 11–6 and the base voltage of Q_1.



Example 11-3

Solution

$V_{REF} = 5.1$ V, the zener voltage. The regulated output voltage is therefore

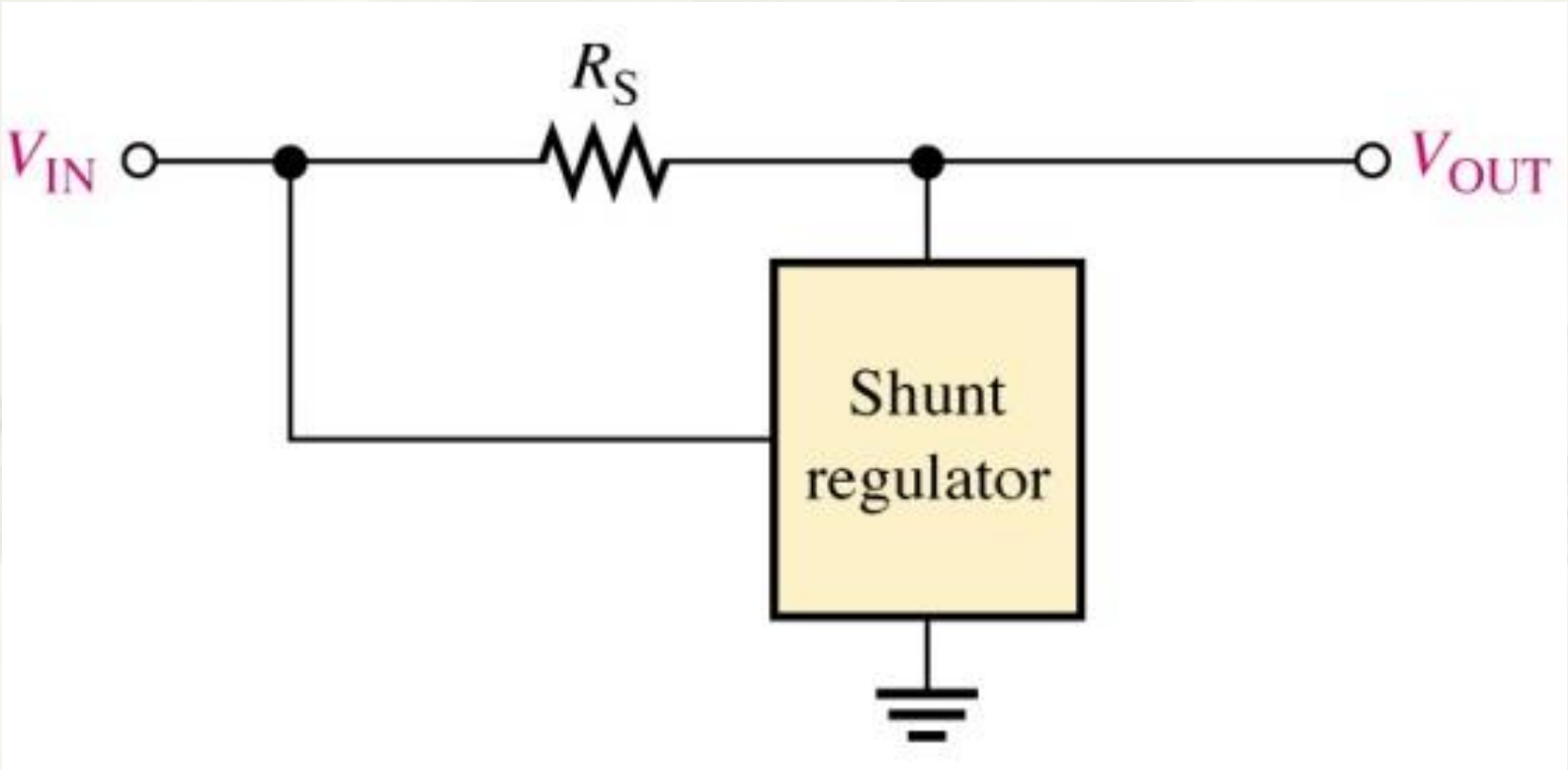
$$V_{OUT} = \left(1 + \frac{R_2}{R_3}\right)V_{REF} = \left(1 + \frac{10\ \text{k}\Omega}{10\ \text{k}\Omega}\right)5.1\ \text{V} = (2)5.1\ \text{V} = \mathbf{10.2\ V}$$

The base voltage of Q_1 is

$$V_B = 10.2\ \text{V} + V_{BE} = 10.2\ \text{V} + 0.7\ \text{V} = \mathbf{10.9\ V}$$

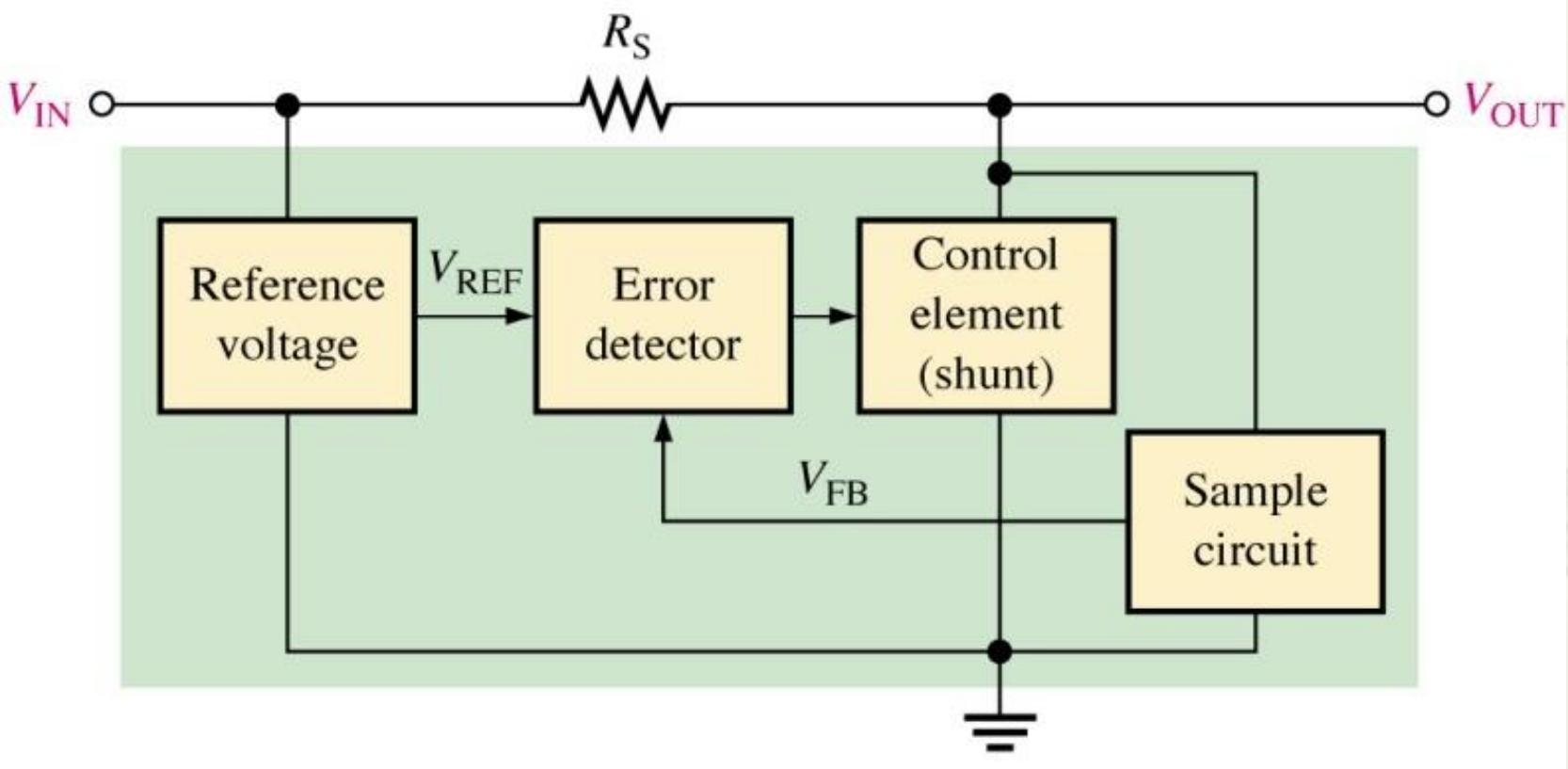
11-3 Linear Shunt Regulators

- **Linear Shunt Regulator**
  - The control element is in parallel (shunt) with the load.



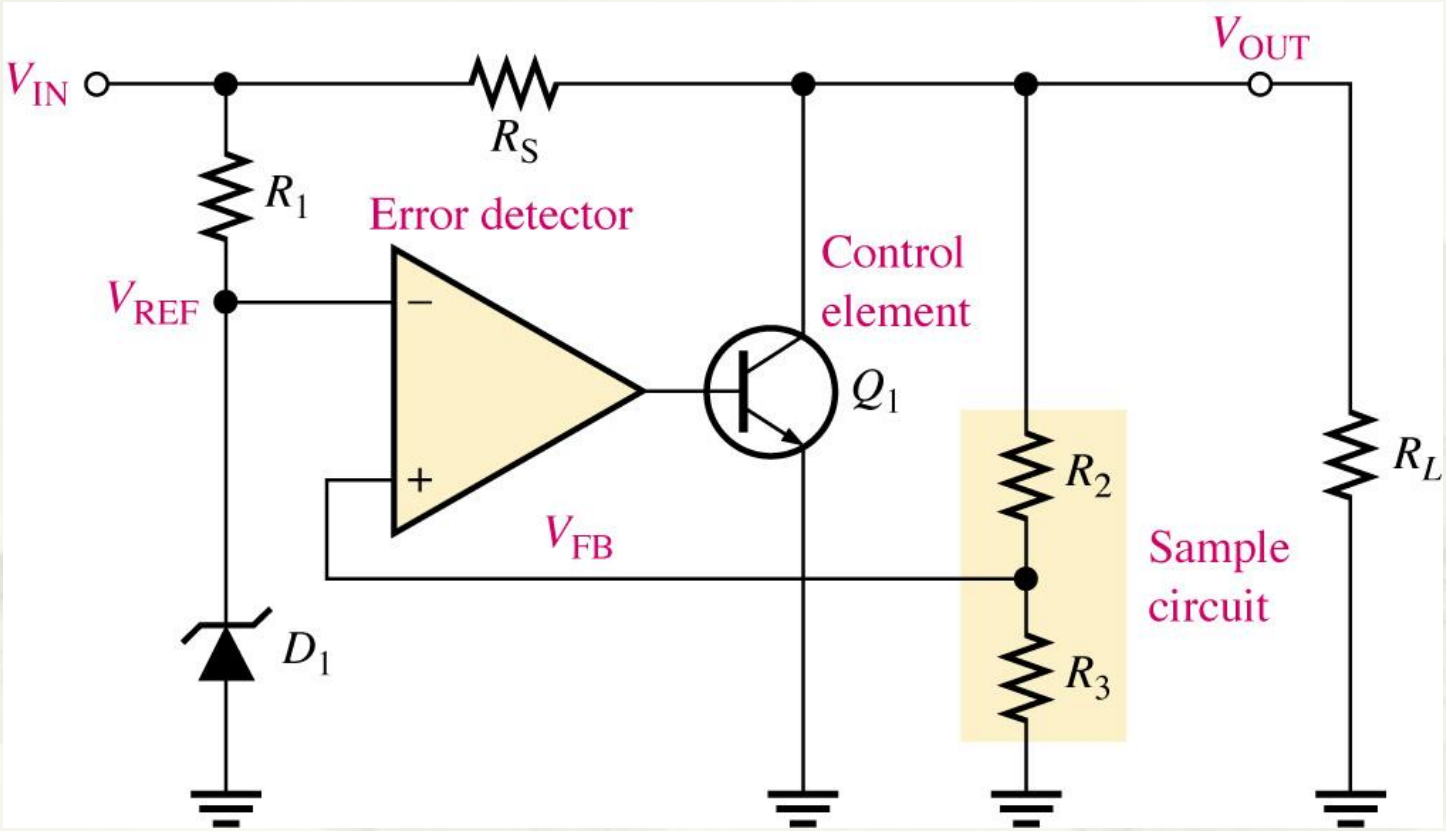
11-3 Linear Shunt Regulators

- **Linear Shunt Regulator**
  - The control element is in parallel (shunt) with the load.



Linear Shunt Regulators

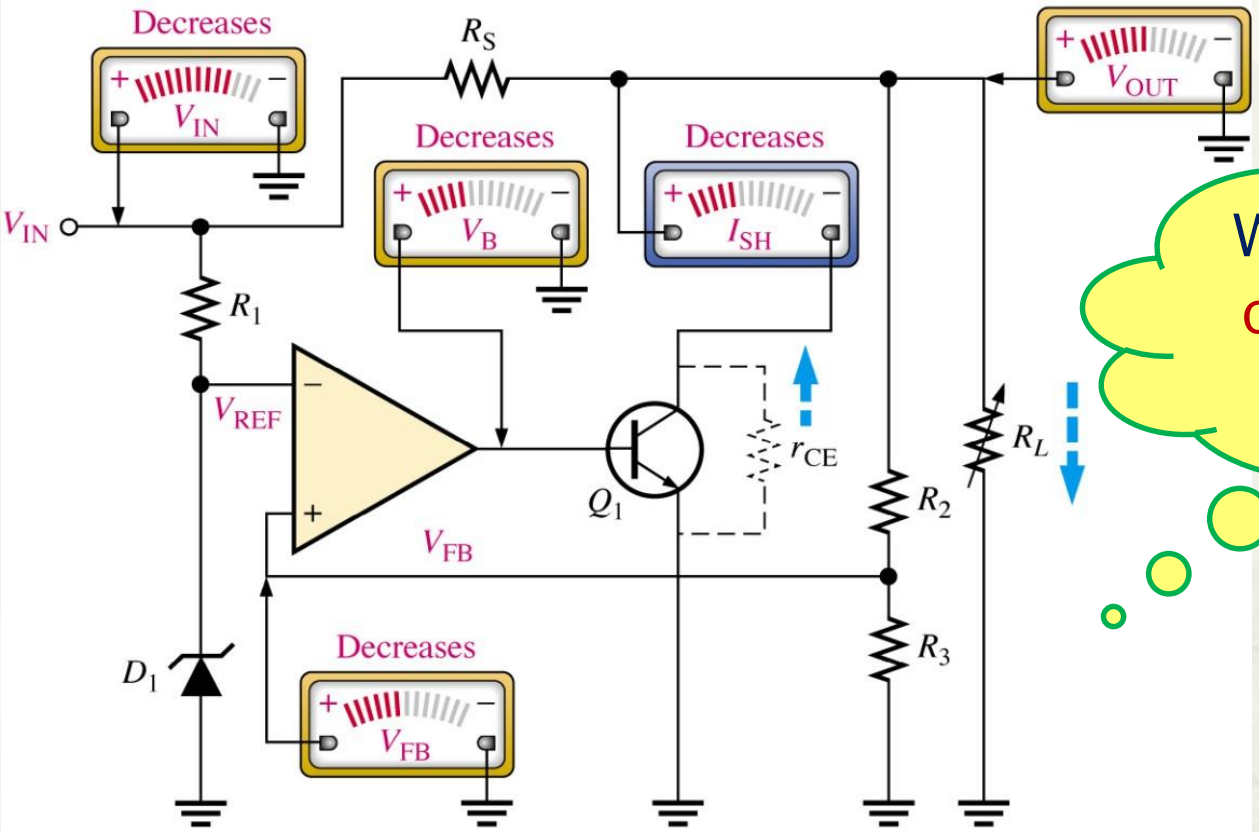
- **Basic op-amp shunt regulator**



$$V_{OUT} = \left(1 + \frac{R_2}{R_3}\right) V_{REF}$$

Linear Shunt Regulators

- **Illustration of shunt regulator action**



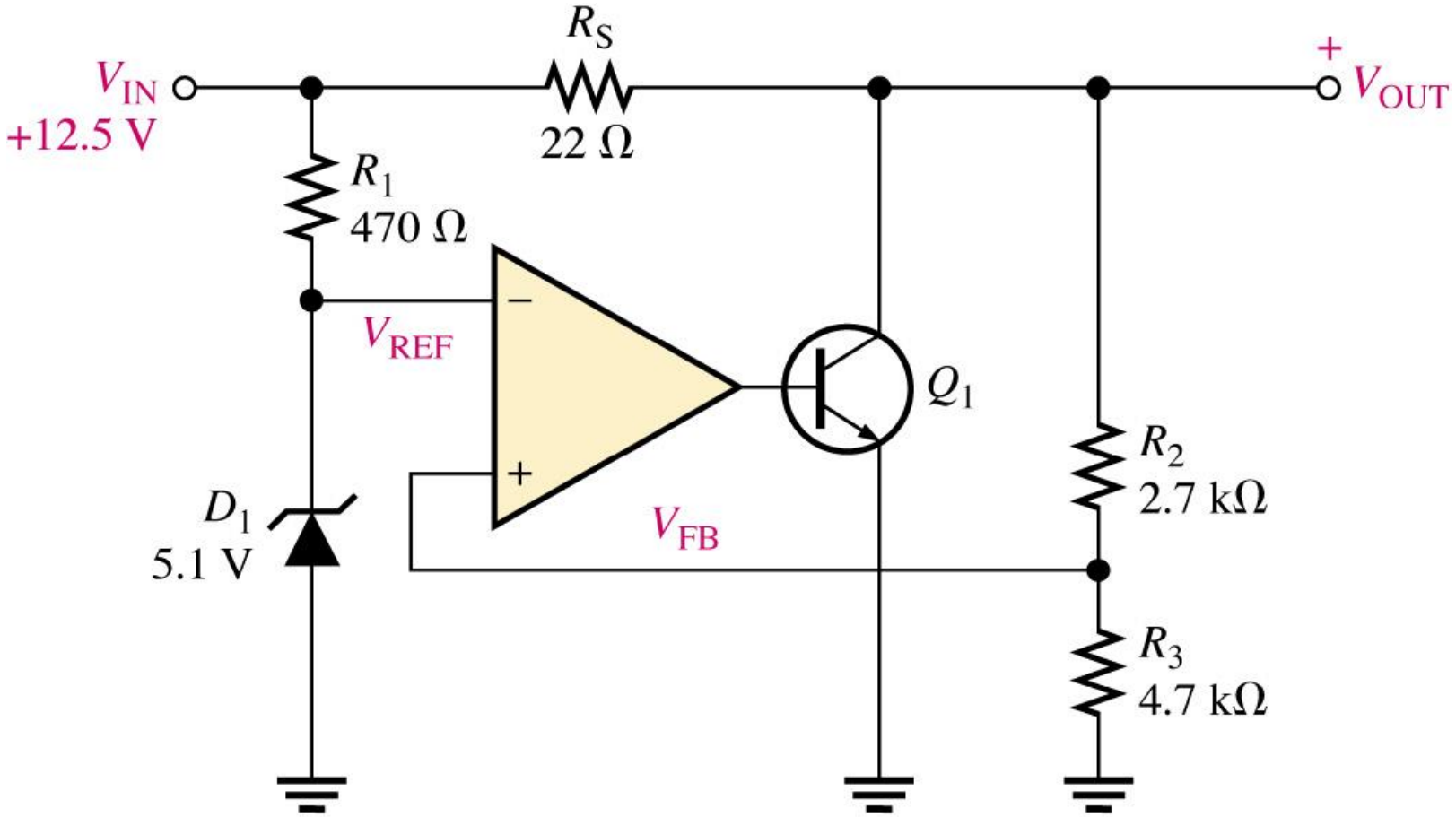
What is the range of output voltage for this circuit?

$$I_{L(max)} = \frac{V_{IN}}{R_S}$$

Example 11-4

Problem

For the circuit in Figure 11–10, what is the output voltage?



Example 11-4

Solution

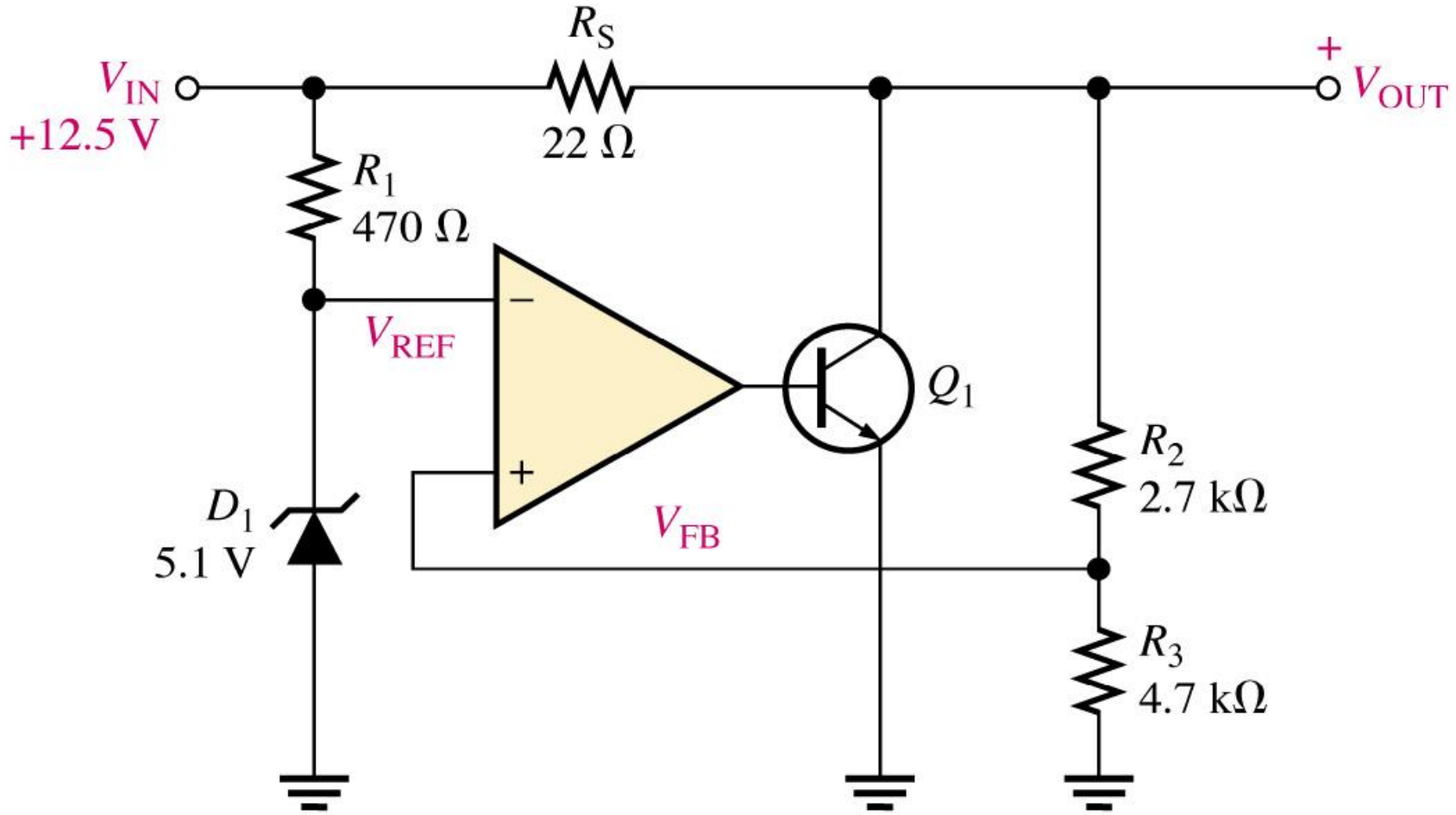
Apply Equation 11–5.

$$V_{OUT} = \left(1 + \frac{R_2}{R_3}\right)V_{REF} = \left(1 + \frac{2.7\,\text{k}\Omega}{4.7\,\text{k}\Omega}\right)5.1\text{ V} = \mathbf{8.03\text{ V}}$$

Example 11-5

Problem

In Figure 11–10, what power rating must R_S have if the maximum input voltage is 12.5 V?



Example 11-5

Solution

The worst-case power dissipation in R_S occurs when the output is short-circuited. $V_{OUT} = 0$, and when $V_{IN} = 12.5$ V, the voltage dropped across R_S is $V_{IN} - V_{OUT} = 12.5$ V. The power dissipation in R_S is

$$P_{R_S} = \frac{V_{R_S}^2}{R_S} = \frac{(12.5\text{ V})^2}{22\ \Omega} = 7.1\text{ W}$$

Therefore, a resistor with at least a **10 W** rating should be used.

The series resistor dissipates a lot of power, so the efficiency is poor.

For higher current requirements, the series regulator is a better choice.

11-4 Switching Regulators

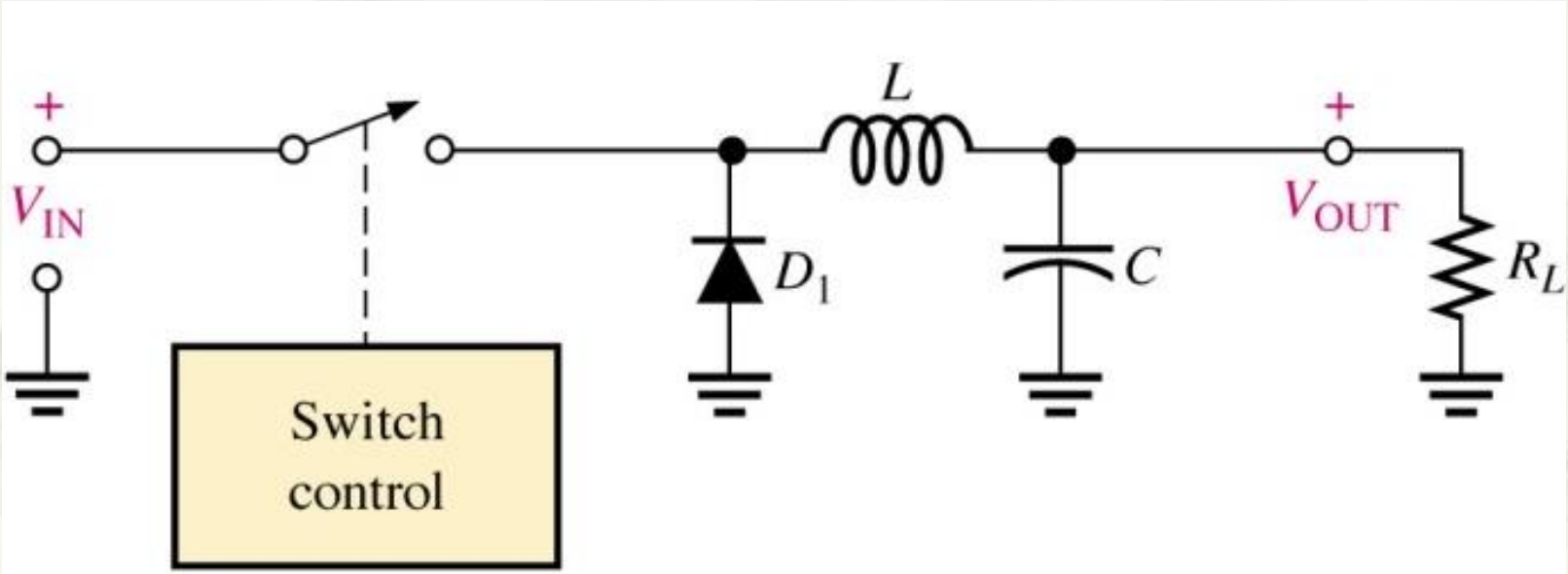
- Step-Down Configuration
- Step-Up Configuration
- Voltage-Inverter Configuration

Switching Regulator

- **Switching regulator**
  - A switching regulator is a very efficient dc to dc converter that can change an unregulated dc input to another voltage, and produce a regulated output.
  - All switching regulators work by turning on and off the input very rapidly creating a pulse train, which is then filtered.
  - SMPS: Switchmode Power Supply
- **Classification**
  - Step-down, step-up, and inverter

Step-Down Configuration

- **Step-down configuration**
  - In the step-down configuration, the output voltage is always less than the input voltage.

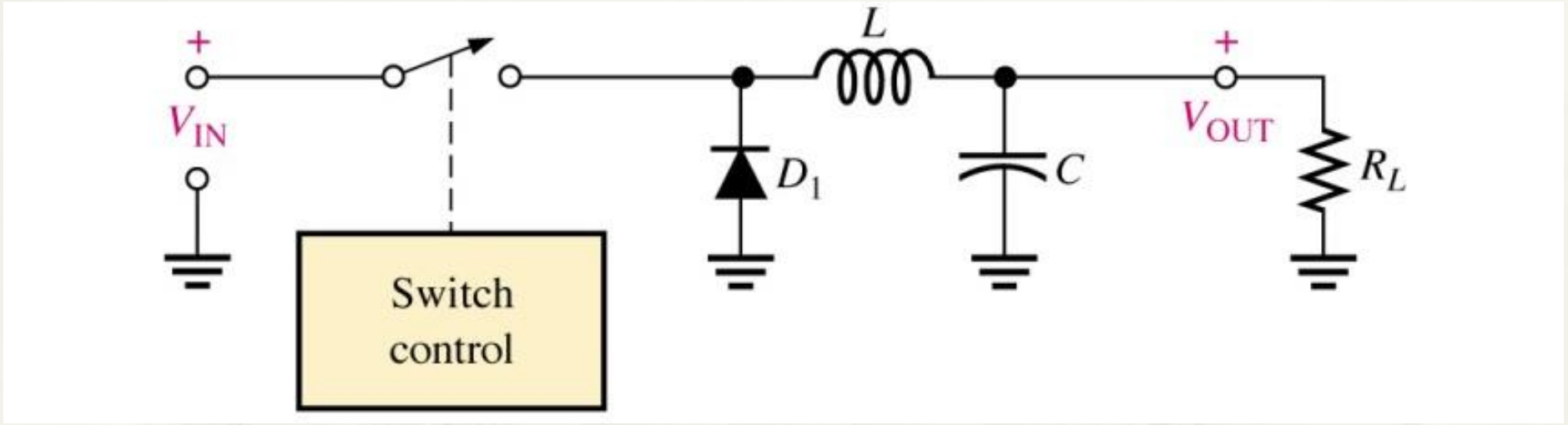


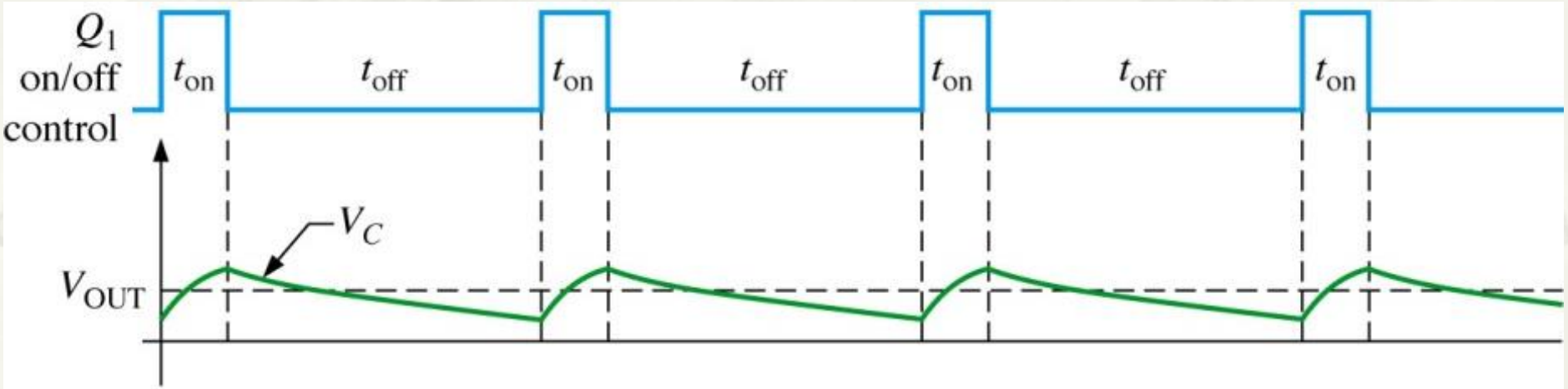
(b) Simplified equivalent circuit

(a) Typical circuit

Also called,
the buck converter.

Switching Regulator Waveforms

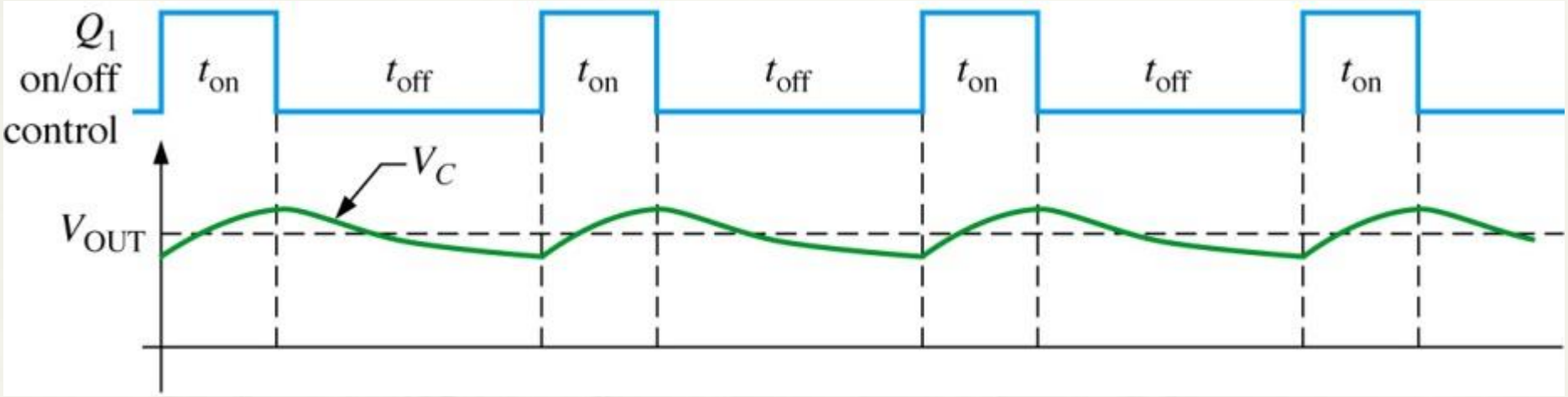




(c) Decrease the duty cycle and V_{OUT} decreases.

(b) Increase the duty cycle and V_{OUT} increases.

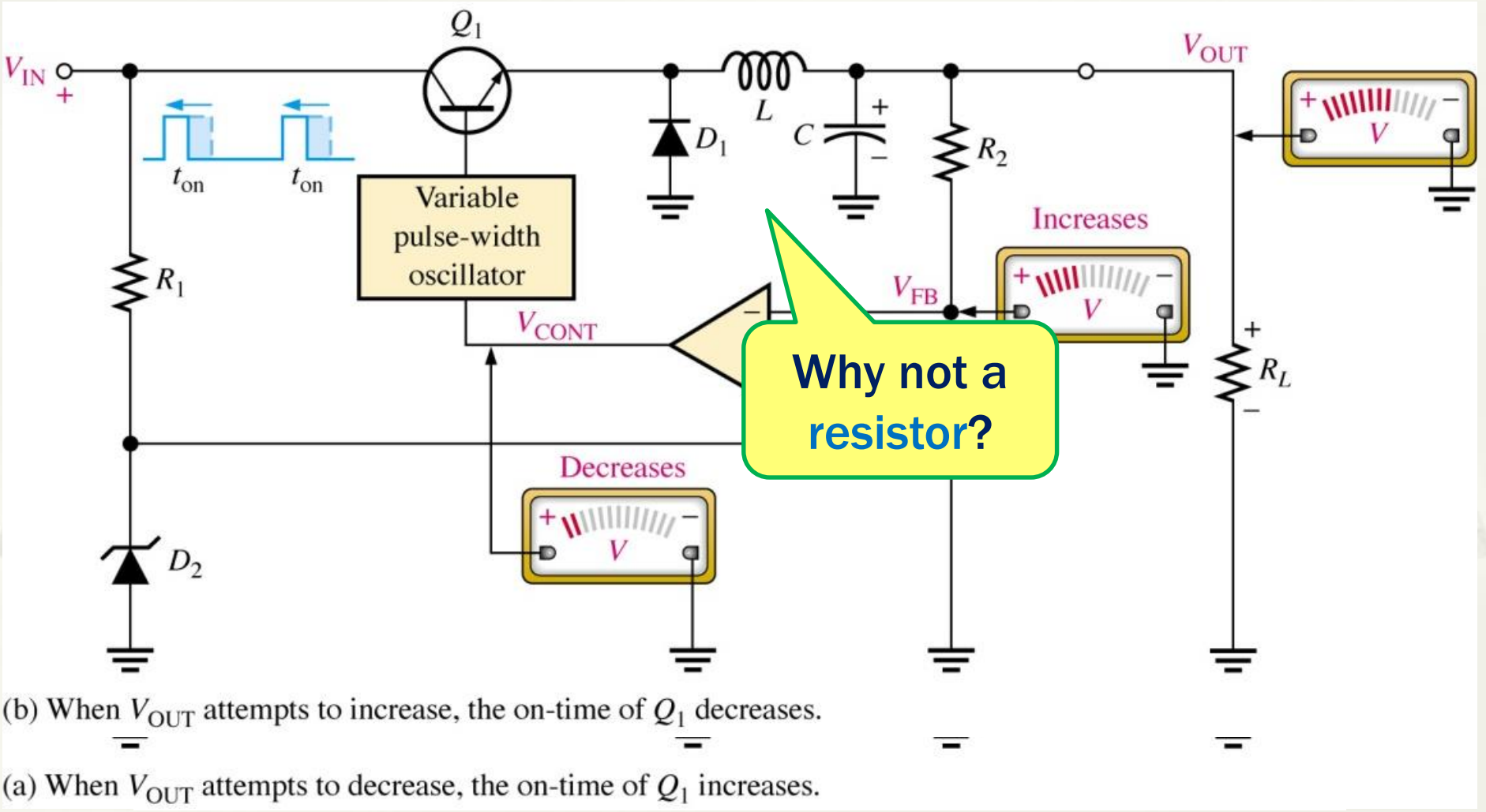
Step-Down Configuration



- **The output voltage**
  - $V_{\mathrm{OUT}} = \left(\frac{t_{\mathrm{on}}}{T}\right) V_{\mathrm{IN}}$, $\frac{t_{\mathrm{on}}}{T}$: the duty cycle
  - $T = t_{\mathrm{on}} + t_{\mathrm{off}}$

Step-Down Configuration

Basic regulating action of a step-down switching regulator

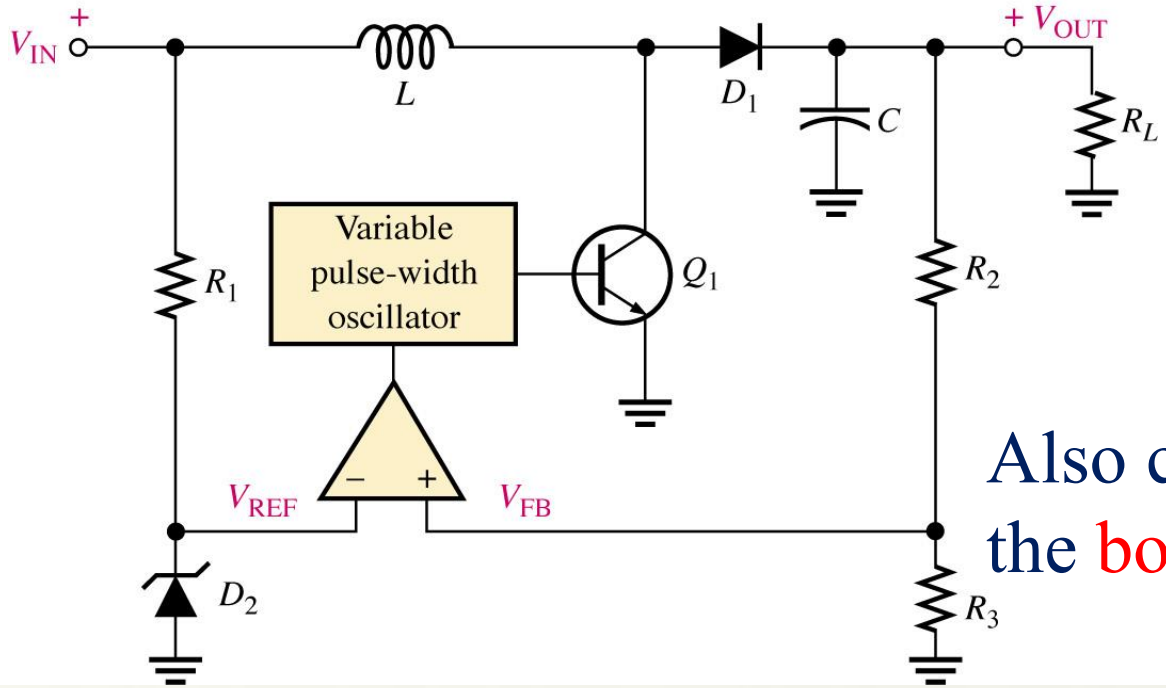


(b) When V_{OUT} attempts to increase, the on-time of Q_1 decreases.

(a) When V_{OUT} attempts to decrease, the on-time of Q_1 increases.

Step-Up Configuration

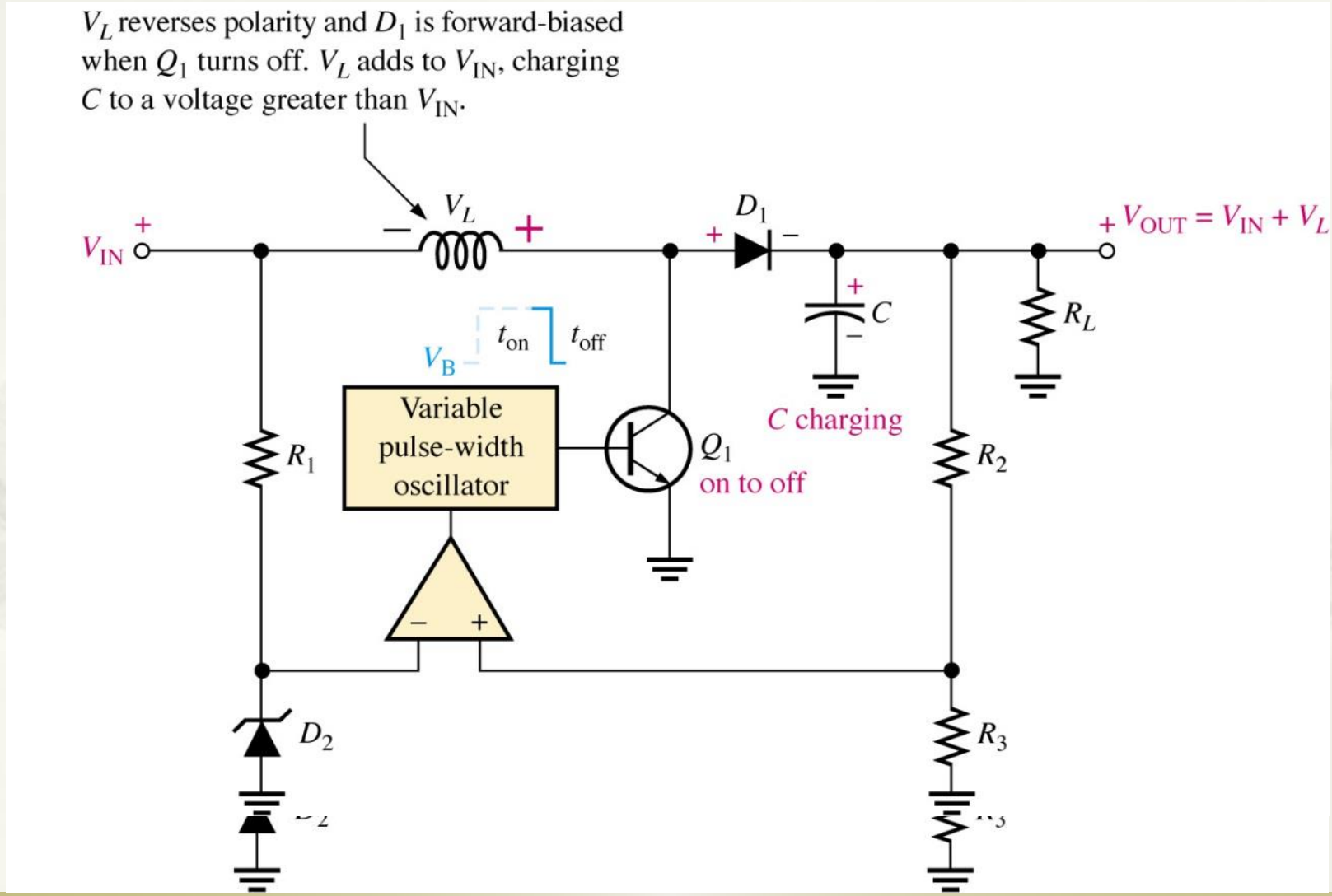
- **Step-up configuration**
  - In the step-up configuration, the output voltage is higher than the input voltage.
  - The transistor operates as a switch to ground.



Also called,
the boost converter.

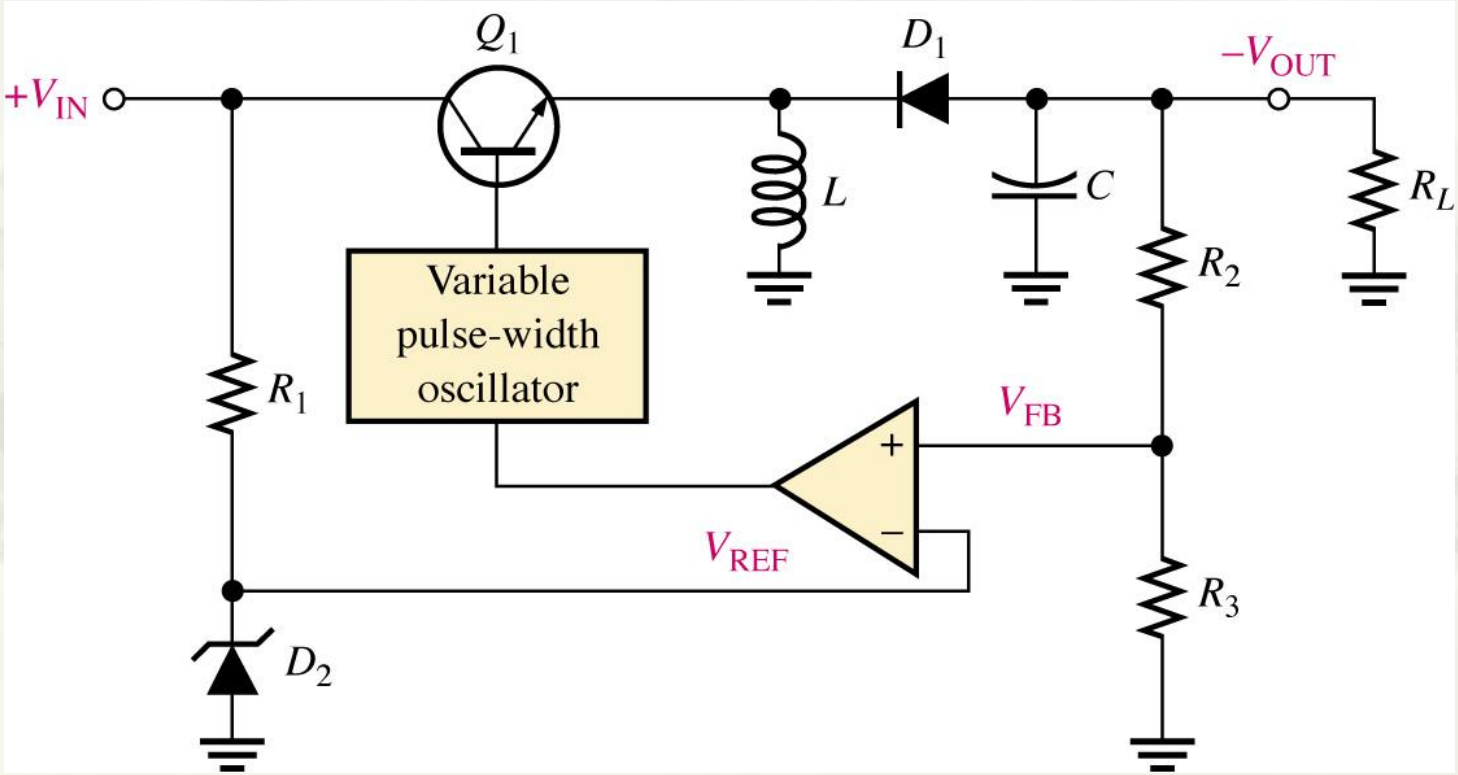
Step-Up Configuration

Basic regulating action of a step-up switching regulator



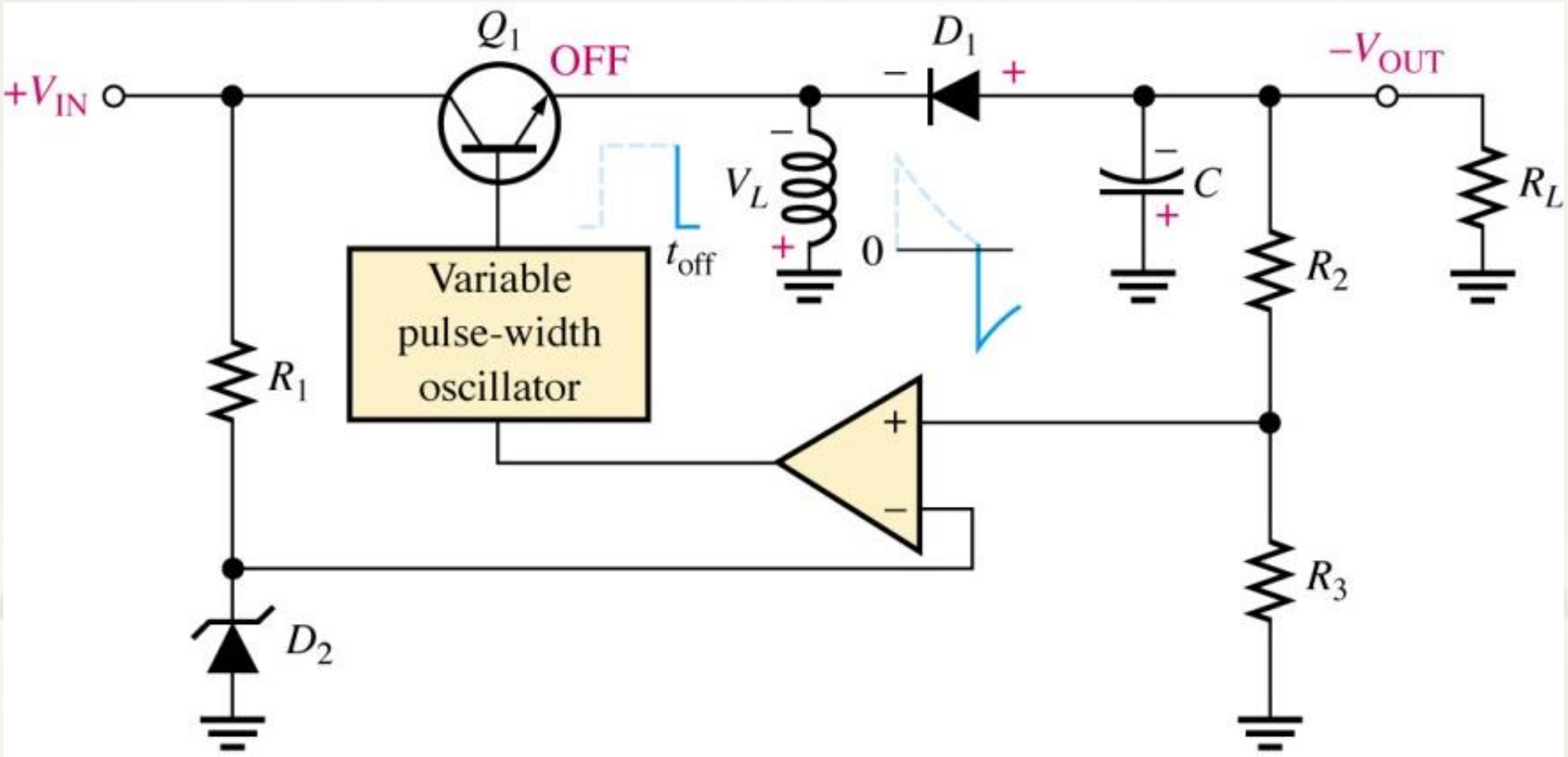
Voltage-Inverter Configuration

- **Voltage-inverter configuration**
  - In the voltage-inverter configuration, the output voltage is opposite in polarity to the input.



Voltage-Inverter Configuration

Basic regulating action of a inverter switching regulator

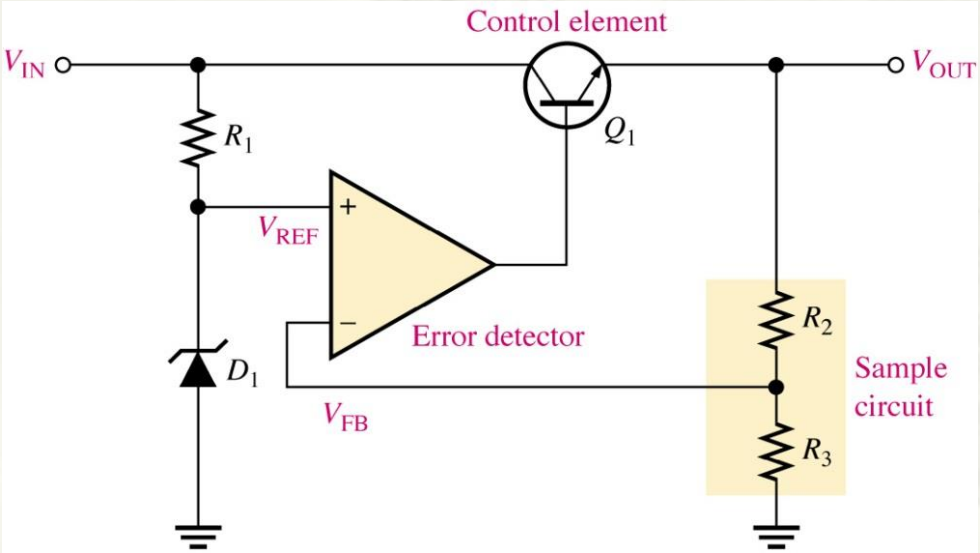


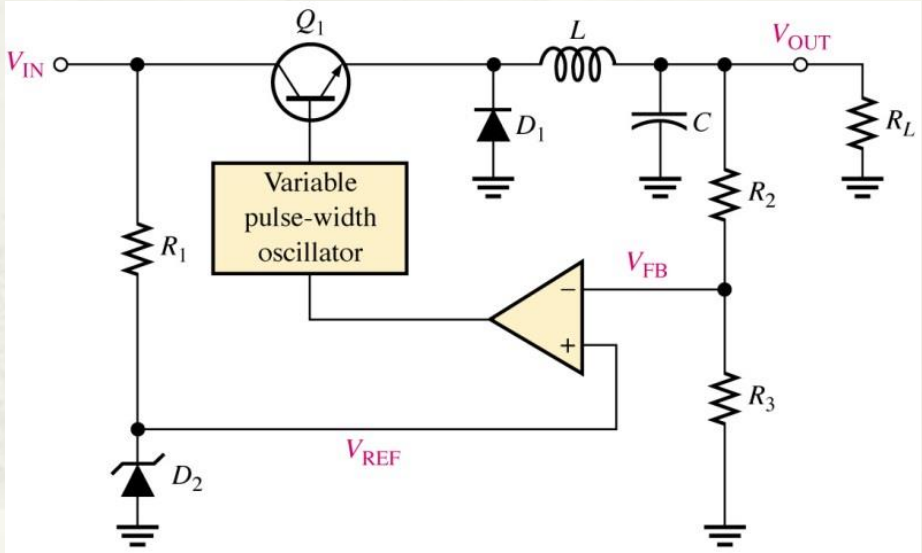
(b) When Q_1 turns off, D_1 is forward-biased.

(a) When Q_1 is on, D_1 is reverse-biased.

The Efficiency of Switching Regulator

- Linear regulator
- Switching regulator



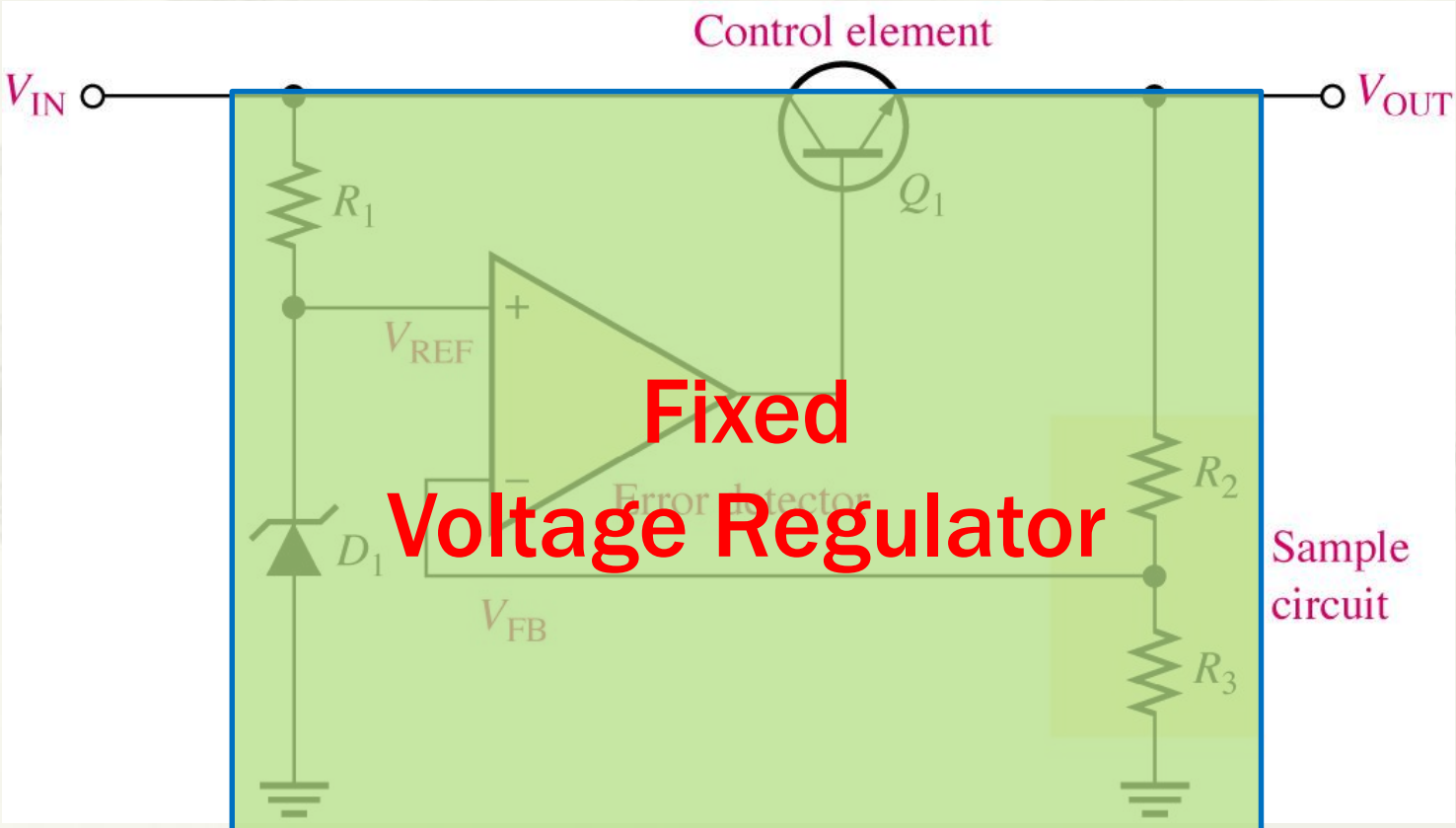


11-5 Integrated Circuit Voltage Regulators

- Fixed Voltage Regulators
- Adjustable Voltage Regulators
- Troubleshooting Three-Terminal Regulators
- Switching Voltage Regulators
- Increasing the Current from an IC Regulator
- A Current Regulator

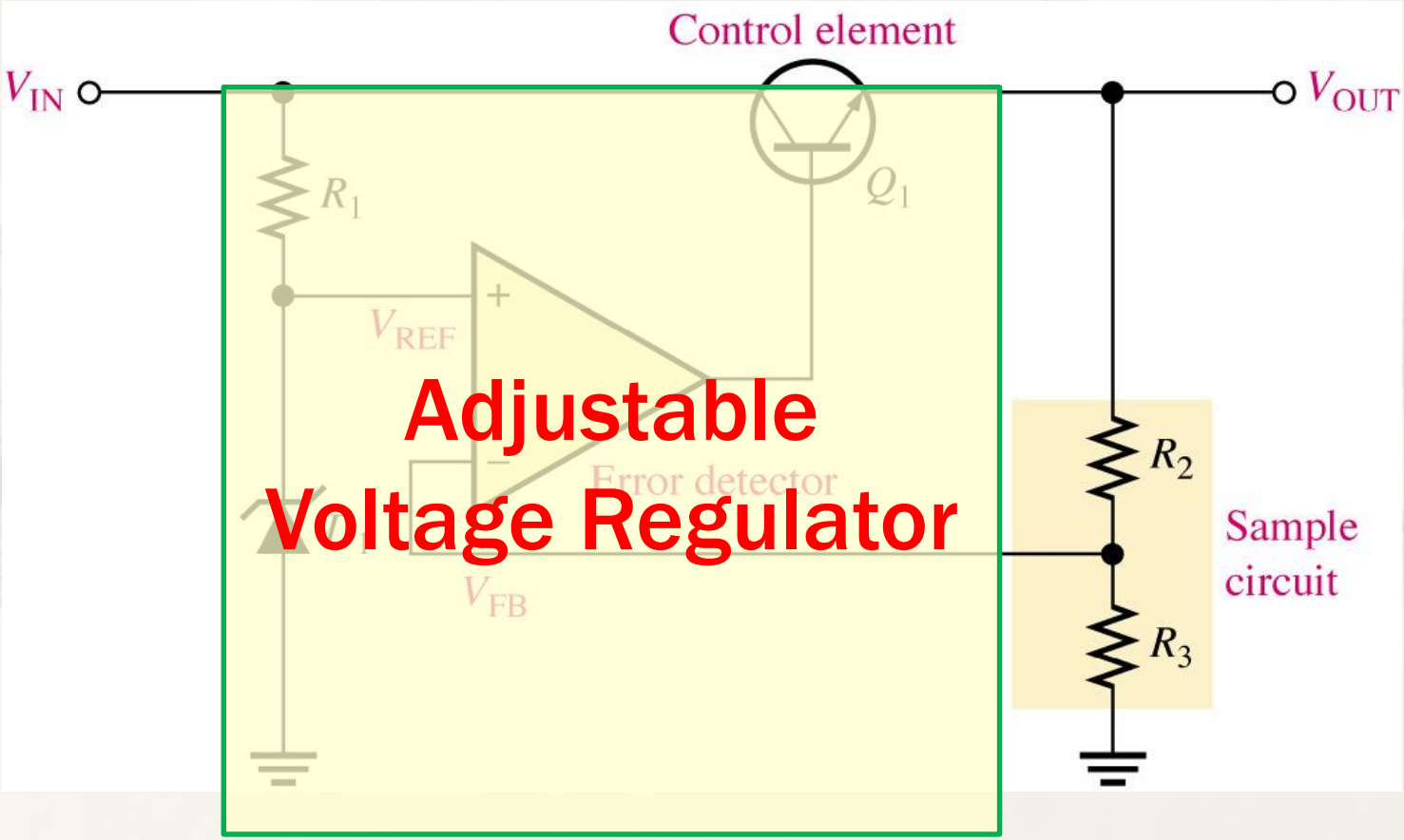
Three-Terminal Regulators

- **The most effective way to remove ripple**
  - a combination of a capacitor-input filter used with an IC regulator
- **Fixed Voltage Regulators**



Three-Terminal Regulators

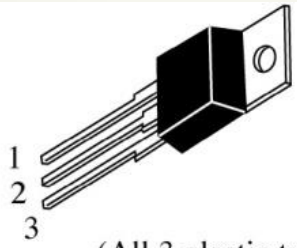
- Fixed Voltage Regulators
- Adjustable Voltage Regulators



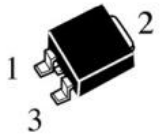
Fixed Voltage Regulators

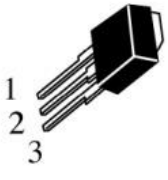
- **Fixed Voltage Regulators: 78XX and 79XX series**

Type number	Output voltage
7805	+5.0 V
7806	+6.0 V
7808	+8.0 V
7809	+9.0 V
7812	+12.0 V
7815	+15.0 V
7818	+18.0 V
7824	+24.0 V

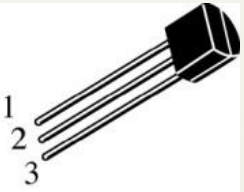


(All 3 plastic types)
Pin 1. Input
2. Ground
3. Output
(Heatsink surface connected to Pin 2)





Pin 1. Output
2. Ground
3. Input

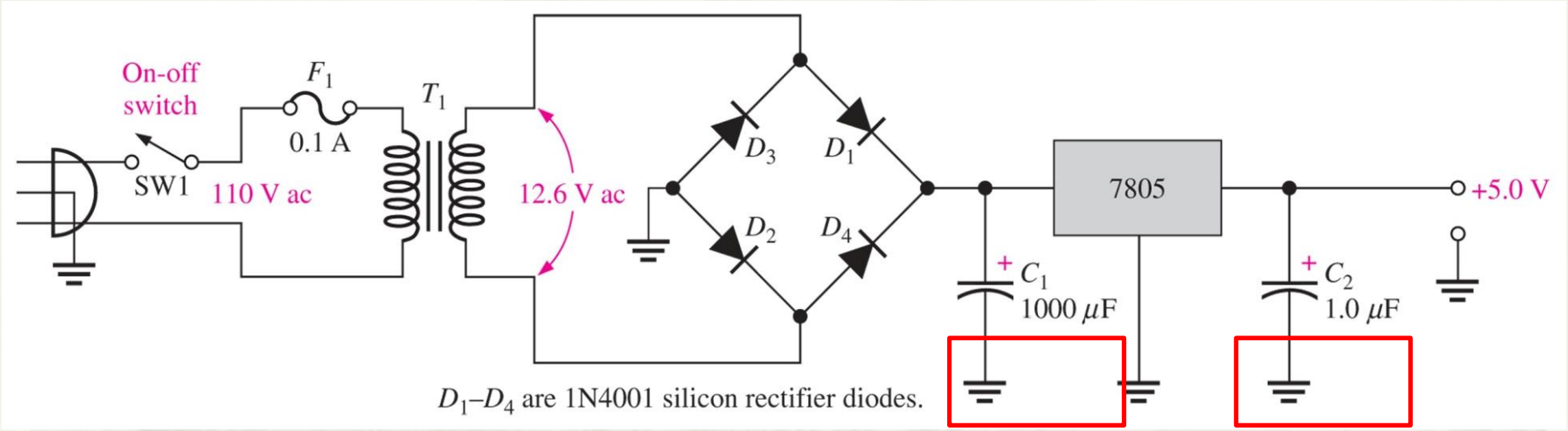




Pin 1. V_{OUT} 5. NC
2. Gnd 6. Gnd
3. Gnd 7. Gnd
4. NC 8. V_{IN}

Fixed Voltage Regulators

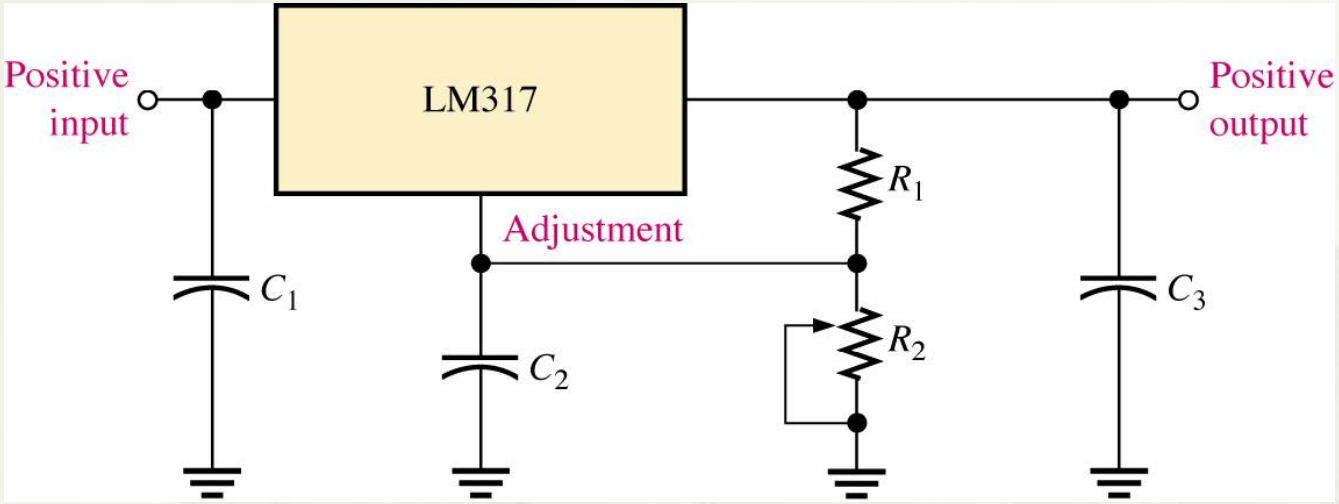
- Fixed positive output voltage: the 7800 series
- Fixed negative output voltage: the 7900 series



Most IC regulators require that the input voltage is at least 2V above the output voltage in order to maintain regulation.

LDO: the Low Dropout Output regulator

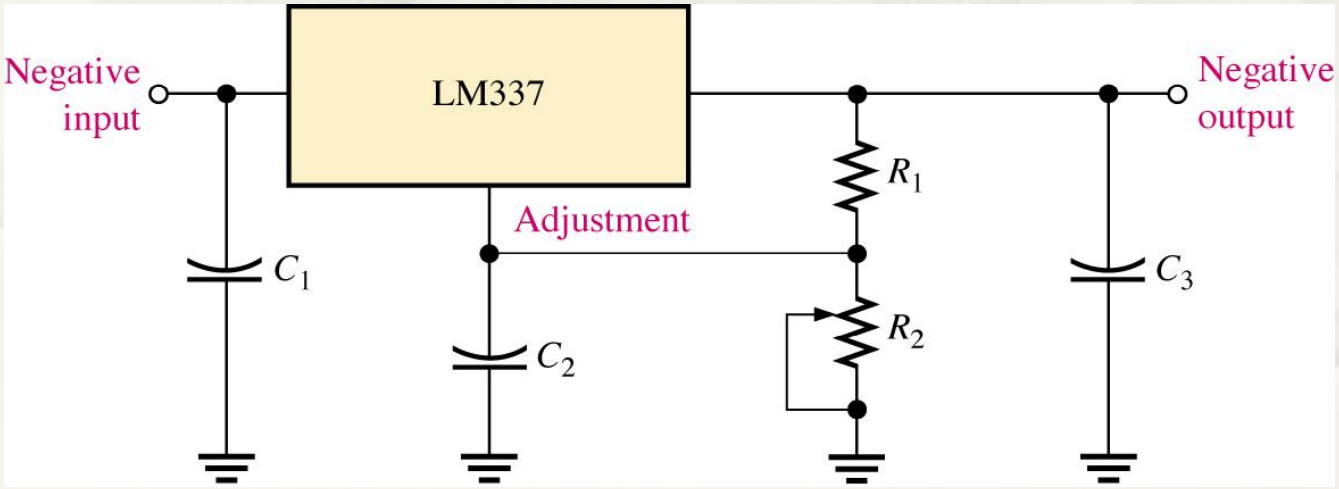
Adjustable Voltage Regulators



Adjustable linear positive voltage regulator LM317

$$V_{out} = 1.25V\left(\frac{R_1 + R_2}{R_1}\right)$$

A basic power supply with a variable output voltage (from 1.25 V to 6.5 V).



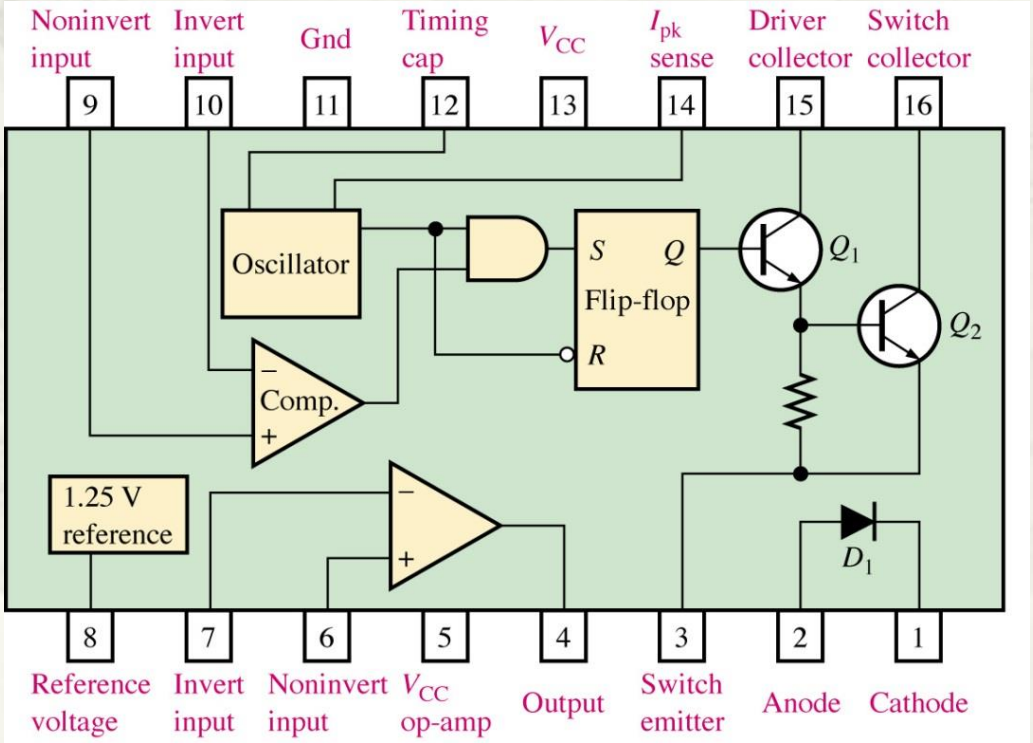
Adjustable linear positive voltage regulator LM337

Troubleshooting Three-Terminal Regulators

- **Three-terminal regulator**
  - Very reliable device
  - When problems occur, the indication is usually an incorrect voltage, high ripple, noisy or oscillating output, or drift.
- **Symptoms**
  - The output voltage is too low
    - ✓ Check the input voltage, the load resistor, the feedback resistors.
  - There is ripple or noise on the output
    - ✓ Check the capacitors.
  - The output is oscillating, has high ripple, or is drifting
    - ✓ Check that the regulator is not too hot or supplying more than its rated current.

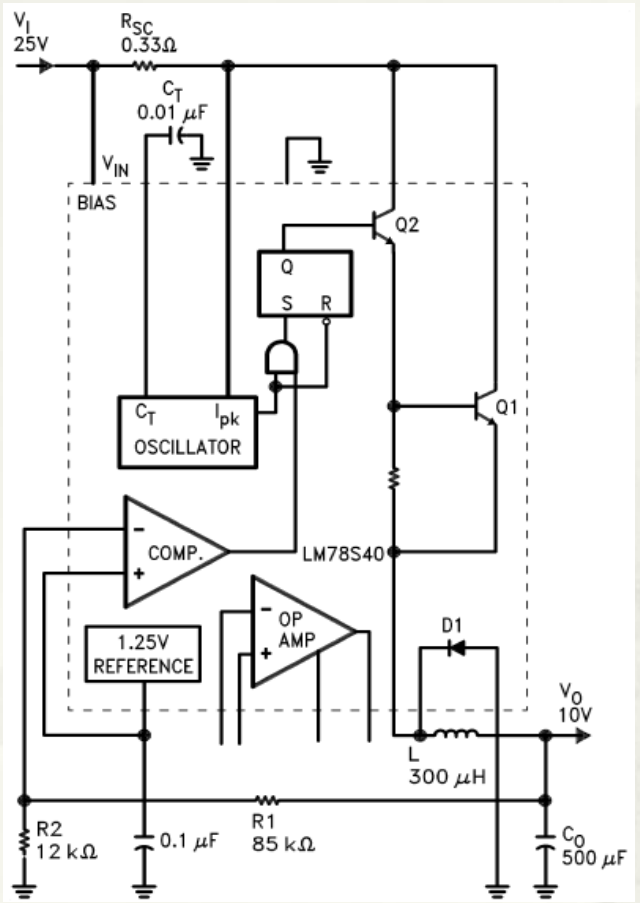
Switching Voltage Regulators

- **Switching voltage regulator IC: 78S40**
  - A universal device that can be used with external components to provide step-up, step-down, and inverting operation.

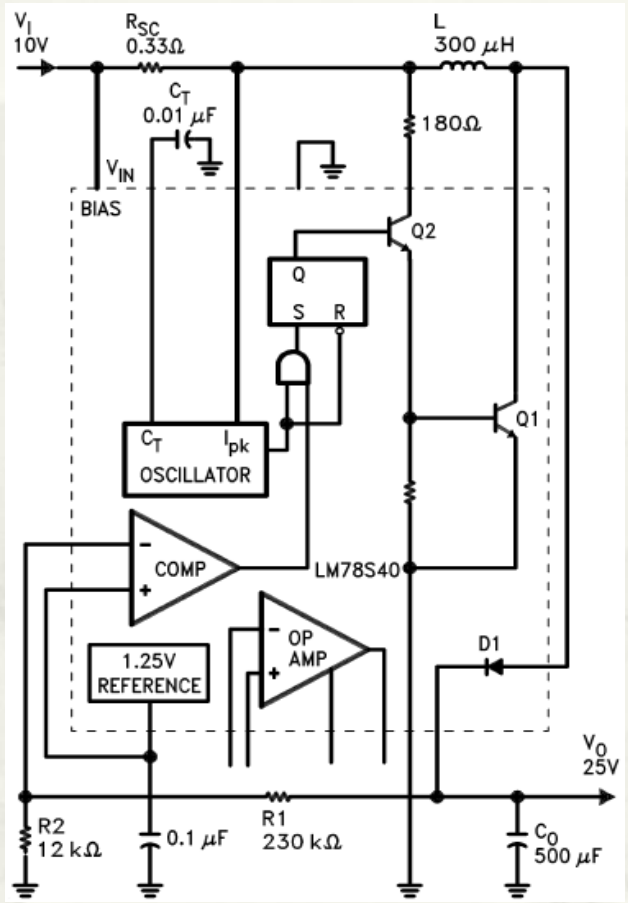


Switching Voltage Regulators

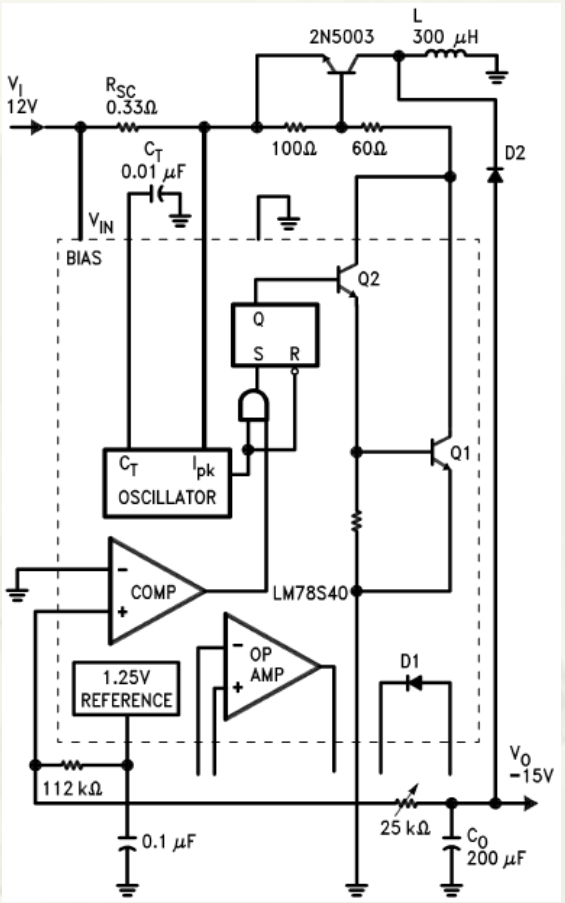
- **Typical applications of 78S40**



Step-down regulator



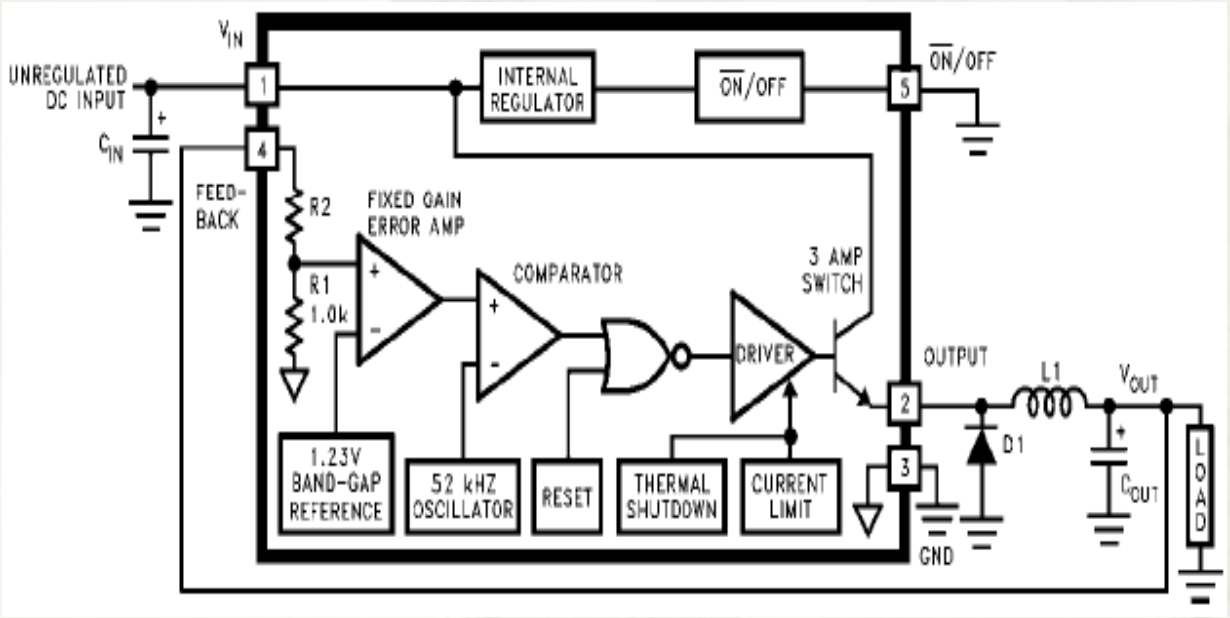
Step-up regulator



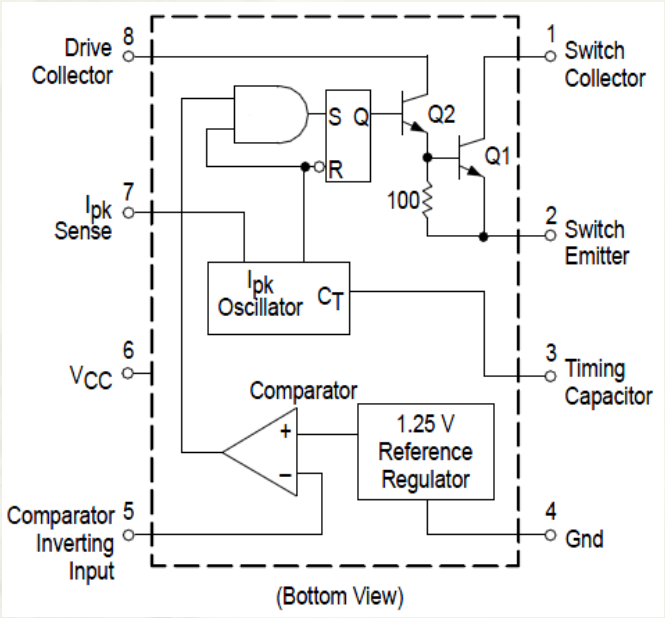
Inverter regulator

Switching Voltage Regulators

- **Other switching voltage regulator IC**
  - Step-Down Voltage Regulator: LM2576
  - DC-to DC Converter Control Circuit: MC34063

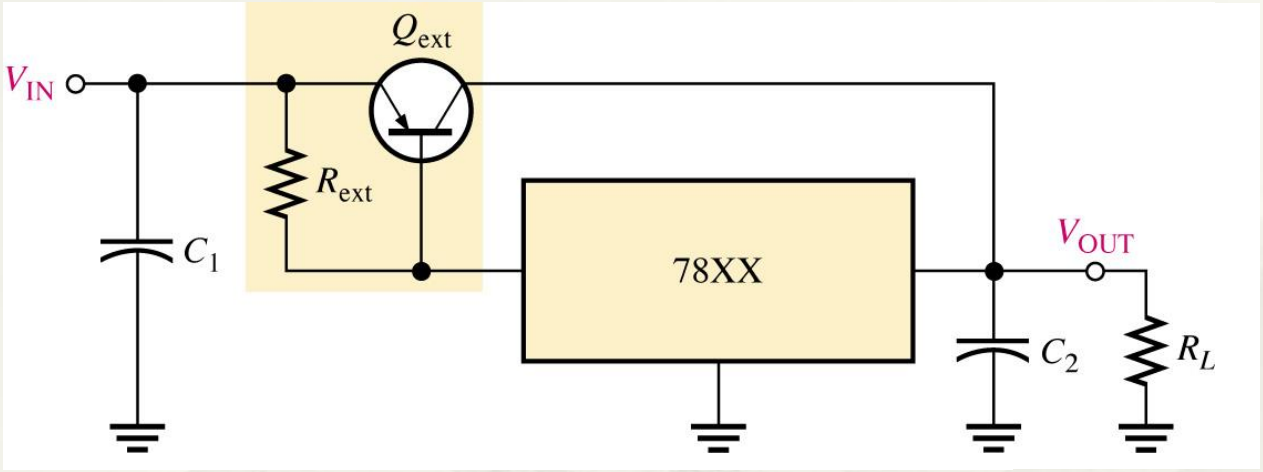


LM2576



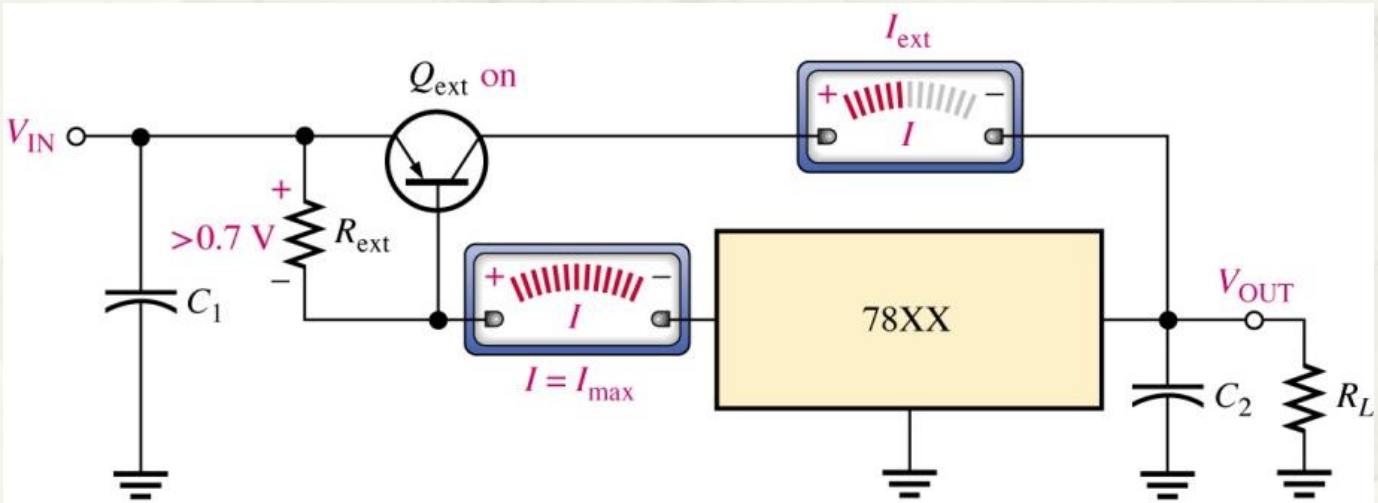
MC34063

Increasing the Current from an IC Regulator



$$R_{ext} = \frac{0.7\text{V}}{I_{max}}$$

$$P_{ext} = I_{ext}(V_{IN} - V_{OUT})$$



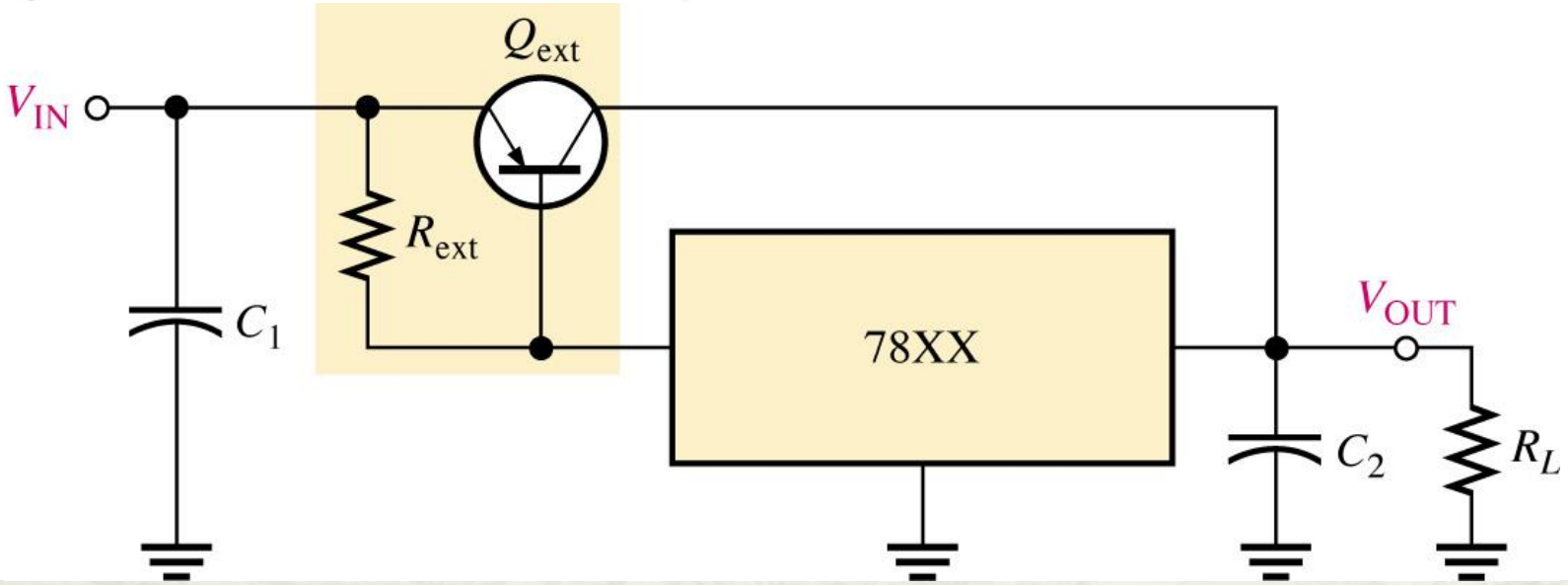
(b) When the load current exceeds I_{max}, the drop across R_{ext} turns Q_{ext} on and the transistor conducts the excess current.

(a) When the regulator current is less than I_{max}, the external pass transistor is off and the regulator is handling all of the current.

Example 11-6

Problem

What value is R_{ext} if the maximum current to be handled internally by the voltage regulator in Figure 11–22 is set at 700 mA?

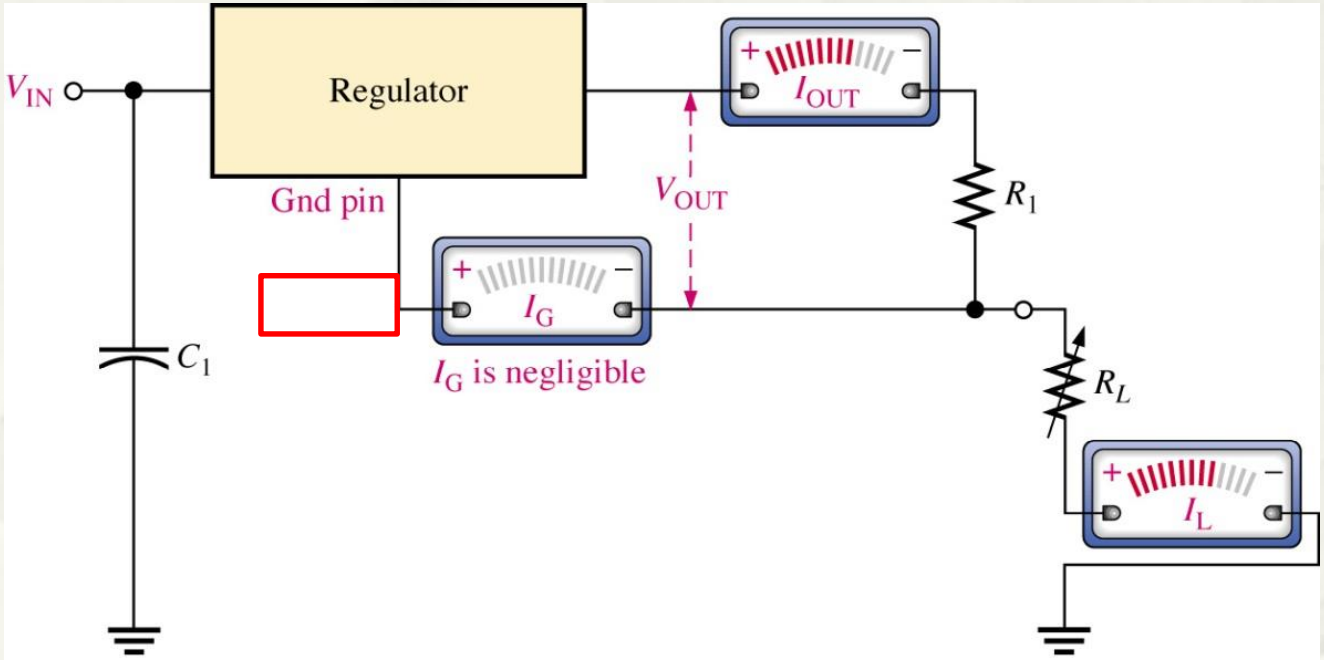


Solution

$$R_{ext} = \frac{0.7\text{ V}}{I_{max}} = \frac{0.7\text{ V}}{0.7\text{ A}} = \mathbf{1\ \Omega}$$

A Current Regulator

- The three-terminal regulator can be used as a current source when an application requires that a constant current be supplied to a variable load.

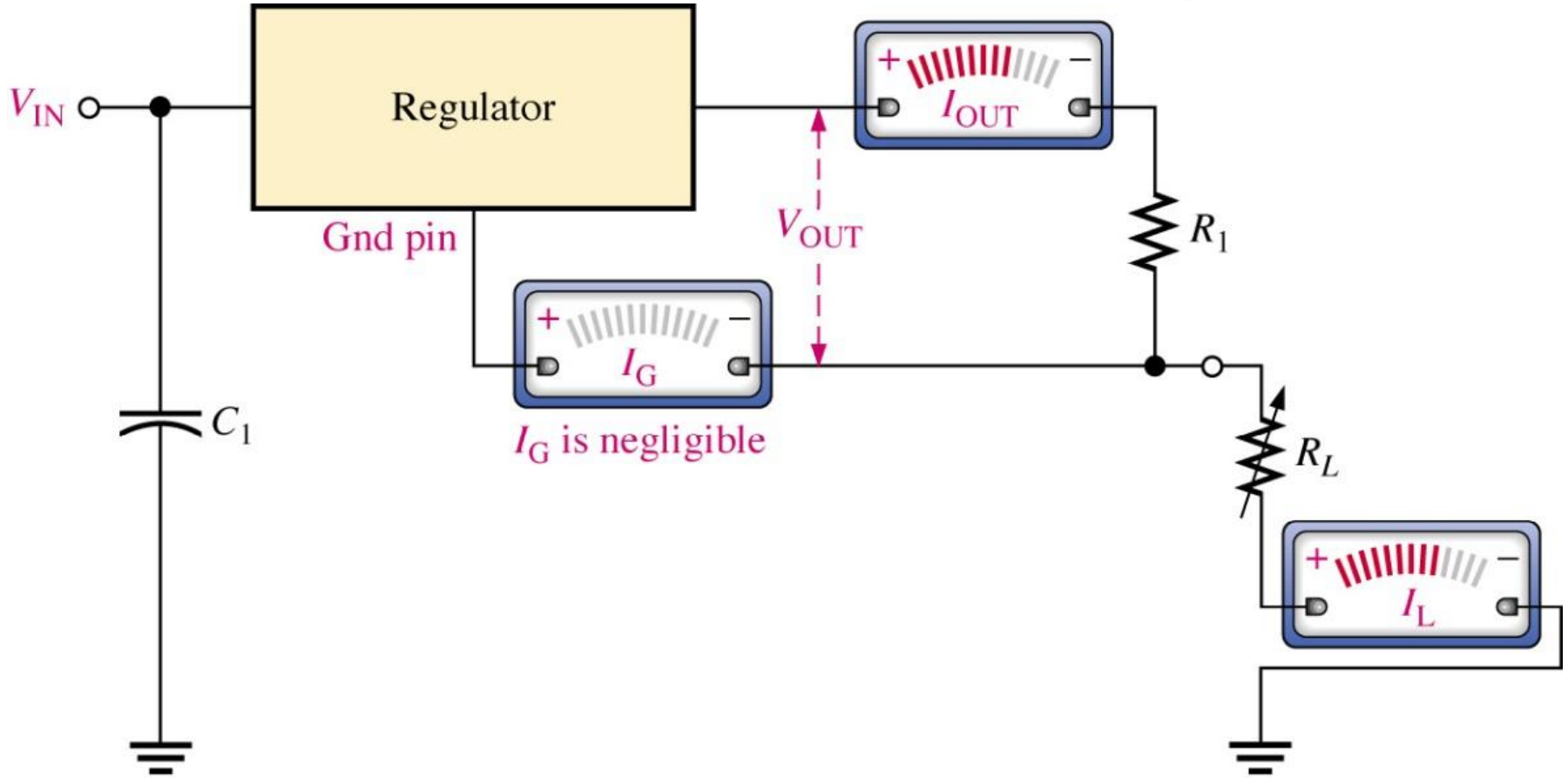


$$I_L = \frac{V_{OUT}}{R_1} + I_G$$

Example 11-7

Problem

What value of R_1 is necessary in a 7805 regulator to provide a constant current of 1 A to a variable load that can be adjusted from 0–10 Ω?



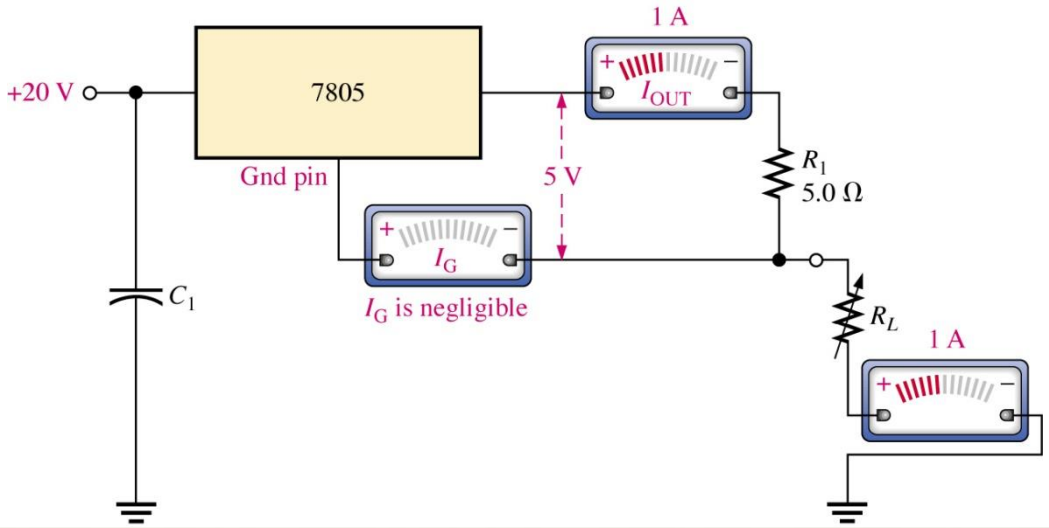
Example 11-7

Solution

First, 1 A is within the limits of the 7805's capability (it can handle at least 1.3 A without an external pass transistor).

The 7805 produces 5 V between its ground terminal and its output terminal. Therefore, if you want 1 A of current, the current-setting resistor must be (neglecting I_G)

$$R_1 = \frac{V_{OUT}}{I_L} = \frac{5\text{ V}}{1\text{ A}} = \mathbf{5.0\ \Omega}$$



KEY OBJECTIVES

1. Given specific voltage parameters, calculate the line and load regulation of a regulator.
2. Discuss the operation of a series load regulator.
3. Discuss the operation of a shunt load regulator.
4. Explain how switching regulators operate.
5. Explain how to configure three-terminal regulators to obtain higher current and how to use them as a current source.

HOMEWORK

- **Chapter Checkup (all)**
  - Just do on your textbook
- **Questions (all)**
  - Odd-numbered in your mind, even-numbered on exercise book
- **Basic Problems**
  - 1, 2, 3, 4, 5, 6, 8, 9, 14, 15
- **Basic-plus Problems**
  - 16, 18, 20, 22